# ON THE SMOOTH TRANSFER OF ORBITAL INTEGRALS FOLLOWING WALDSPURGER

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English translation

*Translation note.* This is a translation of the complete French note dated February 5, 2008. The original section, paragraph, theorem, equation, and reference numbering is retained. Historical statements are translated as written, not updated. Corrections of apparent mathematical or notational slips are identified in translator's footnotes; unresolved ambiguities are retained and, where noted, distinguished from the author's text. Bibliographic titles remain in their published language.

Contents

1. Introduction 1
2. Notation 2
3. Endoscopic data 4
4. Transfer of conjugacy classes 7
5. Harmonic analysis on $p$-adic Lie algebras 10
6. Orbital integrals over a $p$-adic field 11
7. Transfer factors 13
8. Smooth transfer and Fourier transformation: the $p$-adic case 15
9. Orbital integrals on complex Lie algebras 17
10. Pre-stabilization of the trace formula for Lie algebras 18
11. Waldspurger's result 23
References 32

## 1. Introduction

This article is intended as an introduction to Waldspurger's article [9], which proves the existence of transfer assuming the fundamental lemma for Lie algebras, now a theorem. In fact, the existence of transfer is a local problem near semisimple elements. Descent arguments show that it is enough to solve the same problem for the Lie algebra. Waldspurger's approach uses an analogue, for Lie algebras, of the elliptic geometric part of the Arthur–Selberg trace formula. Its spectral counterpart is the Fourier transform of the geometric side, the two being related by the Poisson summation formula. One can then apply the stabilization techniques of Kottwitz and Langlands to compare these trace formulas on the Lie algebras of different groups.

Let us describe the contents of this article in more detail. Section 2 contains the notation used throughout. We recall the notions of abelianized cohomology, for which

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we refer to [5]. Endoscopic data are introduced in Section 3. We have undertaken the Oulipian exercise, dear to Labesse, of avoiding any mention of the Langlands dual group; this accounts for the unusual appearance of the definitions. Unless I am mistaken, their formulation should be equivalent to the usual definition. Section 4 explains how conjugacy classes transfer between a group and its endoscopic group, with particular emphasis on the Kostant section. Sections 5 and 6 recall some facts about $p$-adic harmonic analysis on Lie algebras. In Section 7, we give a definition of transfer factors inspired by Kottwitz's result in [4]. The main statements of endoscopic theory for $p$-adic Lie algebras appear in Section 8: smooth transfer of orbital integrals, compatibility of transfer with Fourier transformation, and the fundamental lemma. The point of Waldspurger's work is to show that each statement follows from the next. Section 9 reviews the analogous theory for complex Lie algebras, whose results rest on those of Bouaziz. Pre-stabilization in our setting is reviewed in Section 10. In the final Section 11, we explain some points in Waldspurger's proof that "the fundamental lemma implies that transfer commutes with the Fourier transform." Consequently, as the title of his article puts it, "the fundamental lemma implies transfer."

## 2. Notation

**2.1.** Let $F$ be a field of characteristic 0. We assume that $F$ is either a $p$-adic field, that is, a finite extension of $\mathbb{Q}_p$, or a number field. Let $\overline{F}$ be an algebraic closure of $F$, and let $\Gamma = \mathrm{Gal}(\overline{F}/F)$ be the associated Galois group.

**2.2.** Let $G$ be a connected reductive group over $F$, and let $\mathfrak{g}$ be its Lie algebra. Let $G_{\mathrm{der}} \subset G$ be its derived subgroup and $G_{\mathrm{sc}}$ the simply connected cover of $G_{\mathrm{der}}$. Let $Z_G$ be the center of $G$ and $G_{\mathrm{ad}} = G/Z_G$ the adjoint group. For any maximal subtorus $T$ of $G$, let $\mathrm{Norm}_G(T)$ be its normalizer in $G$ and $W^G(T) = \mathrm{Norm}_G(T)/T$ the Weyl group. Let $T_{G_{\mathrm{sc}}}$ be the inverse image of $T$ in $G_{\mathrm{sc}}$, and set $T_{G_{\mathrm{ad}}} = T/Z_G$. When the context is clear, we write $T_{\mathrm{sc}} = T_{G_{\mathrm{sc}}}$ and $T_{\mathrm{ad}} = T_{G_{\mathrm{ad}}}$. We will sometimes abuse notation by writing simply $G$ instead of $G(\overline{F})$ for the group of $\overline{F}$-points of $G$.

**2.3.** The group $G$ acts on its Lie algebra by the adjoint action. Let $X \in \mathfrak{g}(F)$. By definition, $X$ is *regular* if its centralizer $G_X$ in $G$ has minimal dimension, equal to the rank of $G$. If $X$ is regular semisimple, its centralizer $G_X$ is a torus. We then write $T_X = G_X$; this is a maximal subtorus of $G$ defined over $F$.

Let $\mathfrak{g}_{\mathrm{reg}}(F)$ be the set of regular semisimple elements of $\mathfrak{g}(F)$. We introduce the set of regular semisimple conjugacy classes

$$\Gamma(\mathfrak{g}) = \mathfrak{g}_{\mathrm{reg}}(F)/G(F).$$

Its stable analogue is

$$\Sigma(\mathfrak{g}) = \mathfrak{g}_{\mathrm{reg}}(F)/G(\overline{F}),$$

namely the quotient of $\mathfrak{g}_{\mathrm{reg}}(F)$ by the following equivalence relation: two elements $X, Y \in \mathfrak{g}_{\mathrm{reg}}(F)$ define the same class in $\Sigma(\mathfrak{g})$ if and only if there is $g \in G$ such that $X = \mathrm{Ad}(g)Y$. We also say that $X$ and $Y$ are *stably conjugate*.

**2.4.** Let $A_G$ be the maximal central split subtorus of $G$. For any torus $T$, the torus $A_T$ is its maximal split subtorus.

Write $\mathfrak{g}_{\mathrm{ell}}(F) \subset \mathfrak{g}_{\mathrm{reg}}(F)$ for the subset of *elliptic* elements: these are the elements $X \in \mathfrak{g}_{\mathrm{reg}}(F)$ whose centralizer $T$ is elliptic, that is, satisfies either of the following equivalent conditions:

– $A_T = A_G$;
– $X_*(T_{G_{\mathrm{sc}}})^\Gamma = 1$.

This notion of ellipticity is preserved by conjugacy and stable conjugacy. The sets $\Gamma_{\mathrm{ell}}(\mathfrak{g})$ and $\Sigma_{\mathrm{ell}}(\mathfrak{g})$ are then defined in the evident way.

**2.5.** In what follows, we abuse notation by not distinguishing a variety defined over $F$ from the variety over $\overline{F}$ obtained by extending scalars. The latter is equipped with an action of $\Gamma$.

Let $G^*$ be a quasi-split inner form of $G$. In other words, $G^*$ is a connected quasi-split reductive group over $F$, and there is an isomorphism over $\overline{F}$

$$\varphi : G \longrightarrow G^* \tag{2.1}$$

such that $\varphi\sigma(\varphi)^{-1}$ is an inner automorphism of $G^*$ for every $\sigma \in \Gamma$. In particular, there is a cocycle $u$ with values in $G^*_{\mathrm{ad}}$ such that

$$\varphi\sigma(\varphi)^{-1} = \mathrm{Ad}(u_\sigma)$$

for every $\sigma \in \Gamma$. The class of $u$ in $H^1(F, G^*_{\mathrm{ad}})$ does not depend on the choice of $\varphi$. Moreover, up to $F$-isomorphism, the groups $G$ that are inner forms of $G^*$ are in natural bijection with $H^1(F, G^*_{\mathrm{ad}})$. The group $G$ is quasi-split if and only if the class of the cocycle $u$ is trivial. The group $G^*$ is defined up to $F$-isomorphism.

**2.6.** Let $T$ be a maximal subtorus of $G$ defined over $F$. The short exact sequence

$$0 \longrightarrow T \longrightarrow G \longrightarrow T\backslash G \longrightarrow 0$$

induces a long exact cohomology sequence

$$\begin{aligned} 0 \longrightarrow H^0(F,T) \longrightarrow H^0(F,G) \longrightarrow H^0(F,T\backslash G) \\ \longrightarrow H^1(F,T) \longrightarrow H^1(F,G). \end{aligned}$$

Following Langlands, Kottwitz, and Labesse, set

$$\begin{aligned} \mathcal{D}(T,G,F) &= \ker\Big(H^1(F,T) \longrightarrow H^1(F,G)\Big) \\ &= \mathrm{coker}\Big(H^0(F,G) \longrightarrow H^0(F,T\backslash G)\Big). \end{aligned}$$

Bear in mind that $H^1(F,G)$ is not a group, but only a set pointed by the class of the trivial cocycle. Let $X \in \mathfrak{g}(F)$ be regular semisimple. The set $\mathcal{D}(T_X, G, F)$ is in bijection with the set of conjugacy classes in the stable conjugacy class of $X$. Recall how this bijection is obtained. Let $Y \in \mathfrak{g}(F)$ be stably conjugate to $X$. There is $g \in G(\overline{F})$ such that $Y = \mathrm{Ad}(g)X$. For every $\sigma \in \Gamma$, one checks that $g^{-1}\sigma(g)$ centralizes $X$, and hence belongs to $T_X$. This gives an element of $\mathcal{D}(T_X, G, F)$.

The following long exact sequence is an "abelianized" version of the preceding one:

$$H^0(F,T) \longrightarrow H^0(F, T_{\mathrm{sc}} \to T) \longrightarrow H^1(F, T_{\mathrm{sc}}) \longrightarrow H^1(F,T) \longrightarrow H^1(F, T_{\mathrm{sc}} \to T).$$

Thus we define

$$\begin{aligned}\mathcal{E}(T,G,F) &= \ker\Big(H^1(F,T) \longrightarrow H^1(F, T_{\mathrm{sc}} \to T)\Big) \\ &= \mathrm{coker}\Big(H^0(F, T_{\mathrm{sc}} \to T) \longrightarrow H^1(F, T_{\mathrm{sc}})\Big).\end{aligned}$$

Recall that Labesse defined cohomology sets $H^i(F, G_{\mathrm{sc}} \to G)$ for the crossed module $[G_{\mathrm{sc}} \to G]$, and that these are isomorphic to $H^i(F, T_{\mathrm{sc}} \to T)$. In degrees $i = 0, 1$, there are abelianization maps

$$H^i(F,G) \longrightarrow H^i(F, G_{\mathrm{sc}} \to G),$$

and hence a map

$$\mathcal{D}(T,G,F) \hookrightarrow \mathcal{E}(T,G,F),$$

which is clearly injective. It is also bijective when $F$ is nonarchimedean local. Indeed, in that case the abelianization map

$$H^1(F,G) \longrightarrow H^1(F, G_{\mathrm{sc}} \to G)$$

is injective: in general, its kernel is the image of $H^1(F, G_{\mathrm{sc}}) \to H^1(F,G)$, but when $F$ is nonarchimedean local it is well known that

$$H^1(F, G_{\mathrm{sc}}) = 1.$$

If $F$ is a number field, we write $\mathbb{A}_F/F$ for the ring of adeles of $F$ and set[1]

$$\mathcal{E}(T,G,\mathbb{A}_F/F) = \mathrm{coker}\Big(H^0(\mathbb{A}_F, T_{\mathrm{sc}} \to T) \longrightarrow H^1(\mathbb{A}_F/F, T_{\mathrm{sc}})\Big).$$

Let $C_F$ be the field $F$ when $F$ is local, or the $F$-algebra of adele classes $\mathbb{A}_F/F$ when $F$ is global. In both cases we define the *group of endoscopic characters*

$$\mathcal{K}(T,G,F) = \mathcal{E}(T,G,C_F)^D,$$

where the superscript $D$ denotes Pontryagin duality.

## 3. Endoscopic data

**3.1. The definition.** Let $G$ be as in the preceding section. An *endoscopic datum* for $G$ is a triple $(H, \kappa, \xi)$, where

– $H$ is a connected quasi-split reductive group over $F$;
– $\xi$ is a commutative diagram

$$\xi: \quad \begin{array}{ccc} T_{H_{\mathrm{sc}}} & \hookrightarrow & T_{G_{\mathrm{sc}}} \\ \downarrow & & \downarrow \\ T_H & \xrightarrow[\iota]{} & T_G. \end{array} \tag{3.1}$$

[1] *Translator's note.* The source calls $\mathbb{A}_F/F$ the "ring of adeles" and subsequently an $F$-algebra of adele classes. These expressions are retained here; the adele ring itself is $\mathbb{A}_F$, whereas its additive quotient by $F$ is not a quotient ring. The notation $H^i(\mathbb{A}_F/F, -)$ is the source's adelic cohomology notation.

Here $T_H$ and $T_G$ are maximal subtori defined over $F$ of $H$ and $G$, respectively; the vertical arrows are the evident maps, the horizontal arrows are $F$-embeddings, and $\iota$ is an $F$-isomorphism;

– $\kappa$ is a character of $H_0(F, X_*(T_{H_{\rm sc}})\backslash X_*(T_{G_{\rm sc}}))$, where $X_*(T_{H_{\rm sc}})$ is identified with its image in $X_*(T_{G_{\rm sc}})$.

These objects must satisfy two conditions, which we now spell out. Let $R_G$ and $R_G^\vee$ be the sets of roots and coroots of $T$ in $G$, respectively. Note that $R_G^\vee$ identifies with a subset of the submodule $X_*(T_{G_{\rm sc}})$ of $X_*(T)$. Set

$$R_\kappa^\vee = \{\alpha^\vee \in R_G^\vee \mid \langle \kappa, \alpha^\vee\rangle = 1\}, \qquad R_\kappa = \{\alpha \in R_G \mid \alpha^\vee \in R_\kappa^\vee\},$$

where $\alpha^\vee$ is the coroot of $\alpha$.

The first condition is as follows. The isomorphism $\iota$ in (3.1) induces an $F$-isomorphism of root data

$$\begin{aligned} &\Big(X^*(T_H), R_H, X_*(T_H), R_H^\vee\Big) \\ &\simeq \Big(X^*(T_G), R_\kappa, X_*(T_G), R_\kappa^\vee\Big). \end{aligned} \tag{3.2}$$

To express the second condition, fix a Galois extension $E$ of $F$ that splits the tori $T_{G_{\rm sc}}$ and $T_{H_{\rm sc}}$. We can then regard $\kappa$ as a character of $H_0(E/F, X_*(T_{H_{\rm sc}})\backslash X_*(T_{G_{\rm sc}}))$. Composing the evident map

$$\begin{aligned} &\widehat{H}^{-1}\Big(E/F, X_*(T_{H_{\rm sc}})\backslash X_*(T_{G_{\rm sc}})\Big) \\ &\longrightarrow H_0\Big(E/F, X_*(T_{H_{\rm sc}})\backslash X_*(T_{G_{\rm sc}})\Big) \end{aligned}$$

with the Tate–Nakayama isomorphism

$$H^1(C_F, T_{H_{\rm sc}} \to T_{G_{\rm sc}}) \simeq \widehat{H}^{-1}\Big(E/F, X_*(T_{H_{\rm sc}})\backslash X_*(T_{G_{\rm sc}})\Big),$$

we see that $\kappa$ induces a character on $H^1(C_F, T_{H_{\rm sc}} \to T_{G_{\rm sc}})$, and hence a character $\kappa_0$ on $H^1(C_F, T_{G_{\rm sc}})$ through the map

$$H^1(C_F, T_{G_{\rm sc}}) \longrightarrow H^1(C_F, T_{H_{\rm sc}} \to T_{G_{\rm sc}}).$$

The second condition concerns this character $\kappa_0$. We require that $\kappa_0$ belong to $\mathcal{K}(T_G, G, F)$, that is, that it be trivial on the image of $H^0(\mathbb{A}_F, T_{G_{\rm sc}} \to T_G)$ if $F$ is global, or of $H^0(F, T_{G_{\rm sc}} \to T_G)$ if $F$ is local.

**3.2. Equivalence.** Let $(H, \kappa, \xi)$ and $(H', \kappa', \xi')$ be two endoscopic data for $G$. Thus we have two commutative diagrams

$$\xi: \quad \begin{array}{ccc} T_{H_{\rm sc}} & \hookrightarrow & T_{G_{\rm sc}} \\ \downarrow & & \downarrow \\ T_H & \xrightarrow[\iota]{} & T_G \end{array} \qquad \text{and} \qquad \xi': \quad \begin{array}{ccc} T_{H'_{\rm sc}} & \hookrightarrow & T'_{G_{\rm sc}} \\ \downarrow & & \downarrow \\ T_{H'} & \xrightarrow[\iota]{} & T'_G. \end{array}$$

By definition, these data are *equivalent* if there is an $F$-isomorphism $\alpha : H' \to H$ and elements $h \in H_{\rm sc}$ and $g \in G_{\rm sc}$ satisfying the following two properties.

(1) We have $(\mathrm{Ad}(h)\circ\alpha)(T_{H'}) = T_H$ and $\mathrm{Ad}(g)(T'_G) = T_G$. The maps $\mathrm{Ad}(h)\circ\alpha$ and $\mathrm{Ad}(g)$ induce an isomorphism over $\overline{F}$ from $\xi'$ to $\xi$. In particular, we have commutative diagrams

$$\begin{array}{ccc} T_{H'_{\mathrm{sc}}} & \longrightarrow & T'_{G_{\mathrm{sc}}} \\ {\scriptstyle \mathrm{Ad}(h)\circ\alpha_{\mathrm{sc}}}\downarrow & & \downarrow{\scriptstyle \mathrm{Ad}(g)} \\ T_{H_{\mathrm{sc}}} & \longrightarrow & T_{G_{\mathrm{sc}}} \end{array}$$

where $\alpha_{\mathrm{sc}}$ is the unique isomorphism $H'_{\mathrm{sc}} \to H_{\mathrm{sc}}$ lifting $\alpha$, and

$$\begin{array}{ccc} T_{H'} & \longrightarrow & T'_G \\ {\scriptstyle \mathrm{Ad}(h)\circ\alpha}\downarrow & & \downarrow{\scriptstyle \mathrm{Ad}(g)} \\ T_H & \longrightarrow & T_G, \end{array}$$

compatible in the evident sense with the diagrams $\xi$ and $\xi'$. The first diagram induces an isomorphism

$$X_*(T_{H'_{\mathrm{sc}}})\backslash X_*(T'_{G_{\mathrm{sc}}}) \longrightarrow X_*(T_{H_{\mathrm{sc}}})\backslash X_*(T_{G_{\mathrm{sc}}}). \tag{3.3}$$

Let us show that it is equivariant. It is enough to prove the analogous assertion with $T'_{G_{\mathrm{sc}}}$ and $T_{G_{\mathrm{sc}}}$ replaced by $T'_G$ and $T_G$. Since the latter are $F$-isomorphic to $T_{H'}$ and $T_H$, it suffices to show that the isomorphism

$$X_*(T_{H'_{\mathrm{sc}}})\backslash X_*(T_{H'}) \longrightarrow X_*(T_{H_{\mathrm{sc}}})\backslash X_*(T_H)$$

induced by $\mathrm{Ad}(h)\circ\alpha$ is $\Gamma$-equivariant. Now

$$\mathrm{Ad}(h)\circ\alpha\circ\sigma(\mathrm{Ad}(h)\circ\alpha)^{-1} = \mathrm{Ad}(h\sigma(h)^{-1}),$$

and $h\sigma(h)^{-1}$ belongs to the normalizer of $T_H$ in $H$. It therefore suffices to check that every element of $W^H(T_H)$ acts trivially on $X_*(T_{H_{\mathrm{sc}}})\backslash X_*(T_H)$. It is enough to check this on the generating set of simple reflections $s_\alpha$, where $\alpha$ ranges over the roots of $T_H$ in $H$. This is immediate from the formula, valid for every $\chi \in X_*(T_H)$,

$$s_\alpha(\chi) - \chi = \alpha(\chi)\alpha^\vee,$$

since $\alpha^\vee \in X_*(T_{H_{\mathrm{sc}}})$.[2]

(2) Identify $\kappa'$ with a character of $H_0(F, X_*(T_{H_{\mathrm{sc}}})\backslash X_*(T_{G_{\mathrm{sc}}}))$ by means of the isomorphism

$$H_0\Big(F, X_*(T_{H'_{\mathrm{sc}}})\backslash X_*(T'_{G_{\mathrm{sc}}})\Big) \longrightarrow H_0\Big(F, X_*(T_{H_{\mathrm{sc}}})\backslash X_*(T_{G_{\mathrm{sc}}})\Big)$$

induced by (3.3). Let $\kappa'_0$ be the character of $H^1(C_F, T_{H_{\mathrm{sc}}} \to T_{G_{\mathrm{sc}}})$ induced by $\kappa'$. One checks that $\kappa'_0$ belongs to $\mathcal{K}(T_G, G, F)$.[3] Let $\kappa_0$ be the element of $\mathcal{K}(T_G, G, F)$ induced by $\kappa$. The second condition is the equality $\kappa_0 = \kappa'_0$.

[2] *Translator's note.* The displayed reflection formula is reproduced as printed. With the usual pairing convention its right-hand side has a minus sign; the divisibility used in the argument is unaffected.

[3] *Translator's note.* Here the source uses $\kappa'_0$ both for the character on the displayed hypercohomology group and for its pullback to $H^1(C_F, T_{G_{\mathrm{sc}}})$. The latter is the character belonging to $\mathcal{K}(T_G, G, F)$.

**3.3. Ellipticity.** An endoscopic datum $(H, \kappa, \xi)$ for $G$ is *elliptic* if $\iota(A_H) = A_G$; the inclusion $\supset$ always holds.

## 4. Transfer of conjugacy classes

**4.1.** Let $\overline{F}[\mathfrak{g}]$ be the $\overline{F}$-algebra of regular polynomial functions on $\mathfrak{g}$. It has an evident action of $\Gamma$. Let $\overline{F}[\mathfrak{g}]^G$ be the subalgebra of $G$-invariant functions: it is stable under $\Gamma$ and is a polynomial algebra over $\overline{F}$. The affine space $\mathfrak{c}_G = \mathrm{Spec}(\overline{F}[\mathfrak{g}]^G)$ also has an action of $\Gamma$. Let

$$\chi = \chi_G : \mathfrak{g} \longrightarrow \mathfrak{c}_G$$

be the morphism dual to the inclusion $\overline{F}[\mathfrak{g}]^G \subset \overline{F}[\mathfrak{g}]$.[4] When $X \in \mathfrak{g}$ is regular semisimple, the fiber $\chi^{-1}(\chi(X))$ is the $G$-conjugacy class of $X$. Note that $\chi$ is defined over $F$. When $X \in \mathfrak{g}_{\mathrm{reg}}(F)$, the $F$-points of the fiber $\chi^{-1}(\chi(X))$ form the stable conjugacy class of $X$.

The map $\chi$ induces maps, still denoted by $\chi$, from $\Gamma(\mathfrak{g})$ and $\Sigma(\mathfrak{g})$ to $\mathfrak{c}_G$.

**4.2.** Let $T$ be a maximal subtorus of $G$ and $W = W^G(T)$. The restriction map $\overline{F}[\mathfrak{g}] \to \overline{F}[\mathfrak{t}]$ induces the Chevalley isomorphism

$$\overline{F}[\mathfrak{g}]^G \longrightarrow \overline{F}[\mathfrak{t}]^W$$

onto the algebra of $W$-invariant polynomials on $\mathfrak{t}$. If $T$ is defined over $F$, this isomorphism is $\Gamma$-equivariant.

**4.3. Transfer of conjugacy classes.** Let $(H, \kappa, \xi)$ be an endoscopic datum for $G$. Let $T_H$ and $T_G$ be its underlying maximal $F$-subtori of $H$ and $G$, and let $\iota : T_H \to T_G$ be its underlying $F$-isomorphism. The definition of an endoscopic datum implies that $W^H = W^H(T_H)$ is naturally a subgroup of $W^G = W^G(T_G)$ and that $\iota$ is $W^H$-equivariant. In particular, $\iota$ induces a $\Gamma$-equivariant map

$$\overline{F}[\mathfrak{t}_G]^{W^G} \longrightarrow \overline{F}[\mathfrak{t}_H]^{W^H}.$$

Combining this map with the Chevalley isomorphism, we obtain a $\Gamma$-equivariant morphism of affine schemes

$$\mathfrak{c}_H \longrightarrow \mathfrak{c}_G. \tag{4.1}$$

Let $Y \in \mathfrak{h}$ and $X \in \mathfrak{g}$. We say that $Y$ is an *image* of $X$, and write $Y \to X$, if $\chi_H(Y)$ maps to $\chi_G(X)$ under this morphism. We say that $Y$ is *$G$-regular* if there exists a regular semisimple $X$ such that $Y \to X$, or equivalently if every $X$ such that $Y \to X$ is regular semisimple.

**Definition 4.1.** Let $Y \in \mathfrak{h}(F)$ be $G$-regular. We say that $Y$ *transfers to* $\mathfrak{g}$ if there is $X \in \mathfrak{g}_{\mathrm{reg}}(F)$ such that $Y \to X$.

If $Y \to X$, the same holds for any $Y' \in \mathfrak{h}(F)$ and $X' \in \mathfrak{g}(F)$ such that $Y'$ is $H$-stably conjugate to $Y$ and $X'$ is $G$-stably conjugate to $X$. All $X' \in \mathfrak{g}(F)$ such that $Y \to X'$ are stably conjugate. Let $\Sigma_G(\mathfrak{h}) \subset \Sigma(\mathfrak{h})$ be the subset of $G$-regular classes. We have thus defined a function

$$\Sigma_G(\mathfrak{h}) \longrightarrow \Sigma(\mathfrak{g}),$$

[4] *Translator's note.* The source repeats the superscript $G$ on the right of this inclusion. It also writes $\gamma(\mathfrak{g})$ instead of $\Gamma(\mathfrak{g})$ at the end of this paragraph. Both notational slips are corrected here.

whose domain consists of the classes that are images of elements of $\mathfrak{g}_{\mathrm{reg}}(F)$, and whose fibers are readily checked to be finite.

**4.4.** Let $Y \in \mathfrak{h}(F)$ be $G$-regular and $X \in \mathfrak{g}_{\mathrm{reg}}(F)$. One checks that $Y \to X$ if and only if there are $h \in H_{\mathrm{sc}}$ and $g \in G_{\mathrm{sc}}$ such that

– $\mathrm{Ad}(h)T_Y = T_H$;
– $\mathrm{Ad}(g)T_G = T_X$;
– the composite morphism

$$T_Y \xrightarrow{\mathrm{Ad}(h)} T_H \xrightarrow{\iota} T_G \xrightarrow{\mathrm{Ad}(g)} T_X \tag{4.2}$$

is an $F$-isomorphism carrying $Y$ to $X$.

The isomorphism (4.2) extends in the evident way to a diagram

$$\xi' : \quad \begin{array}{ccc} T_{Y,H_{\mathrm{sc}}} & \hookrightarrow & T_{X,G_{\mathrm{sc}}} \\ \downarrow & & \downarrow \\ T_Y & \longrightarrow & T_X. \end{array}$$

The inner automorphisms $\mathrm{Ad}(h)$ and $\mathrm{Ad}(g^{-1})$ induce an isomorphism from $\xi'$ to $\xi$. As we saw after (3.3), this implies that $\mathrm{Ad}(g^{-1})$ induces a $\Gamma$-equivariant isomorphism

$$X_*(T_{Y,H_{\mathrm{sc}}})\backslash X_*(T_{X,G_{\mathrm{sc}}}) \longrightarrow X_*(T_{H_{\mathrm{sc}}})\backslash X_*(T_{G_{\mathrm{sc}}}). \tag{4.3}$$

Through this isomorphism, $\kappa$ defines a character $\kappa'$ of $H_0(X_*(T_{Y,H_{\mathrm{sc}}})\backslash X_*(T_{X,G_{\mathrm{sc}}}))$. One checks that $\kappa'$ is endoscopic: it defines on $H^1(C_F, T_{X,G_{\mathrm{sc}}})$ a character $\kappa_Y$ belonging to $\mathcal{K}(T_X, G, F)$. We therefore have an endoscopic datum $(H, \kappa', \xi')$. The identity of $H$ extends to an equivalence between $(H, \kappa, \xi)$ and $(H, \kappa', \xi')$.

**4.5.** Two equivalent endoscopic data $(H, \kappa, \xi)$ and $(H, \kappa', \xi')$ give the same morphism $\mathfrak{c}_H \to \mathfrak{c}_G$ in (4.1), and the same function $\Sigma_G(\mathfrak{h}) \to \Sigma(\mathfrak{g})$.

**4.6.** In this paragraph, assume that $G = G^*$, that is, that $G$ is quasi-split. Let $B$ be a Borel subgroup of $G$ defined over $F$, and let $T$ be a subtorus of $B$ defined over $F$. Let $\Phi$ be the set of simple roots of $T$ in $B$. For every root $\alpha$ of $T$ in $G$, the eigenspace

$$\mathfrak{g}_\alpha = \{X \in \mathfrak{g} \mid \text{for every } t \in T,\ \mathrm{Ad}(t)X = \alpha(t)X\}$$

is one-dimensional over $\overline{F}$. For each $\alpha \in \Phi$, fix a nonzero element $X_\alpha \in \mathfrak{g}_\alpha$ such that $\sigma(X_\alpha) = X_{\sigma(\alpha)}$ for every $\sigma \in \Gamma$. The triple $(B, T, \{X_\alpha\})$ is called a *pinning*.

For every $\alpha \in \Phi$, let $X_{-\alpha}$ be the unique element of $\mathfrak{g}_{-\alpha}$ characterized by the condition that $[X_\alpha, X_{-\alpha}]$ is the coroot $\alpha^\vee$, viewed as an element of $\mathfrak{t}$. Set

$$X_+ = \sum_{\alpha \in \Phi} X_\alpha, \qquad X_- = \sum_{\alpha \in \Phi} X_{-\alpha}.$$

Let $\mathfrak{g}_{X_+}$ be the centralizer of $X_+$ in $\mathfrak{g}$.[5] We state the following theorem of Kostant.

**Theorem 4.2.** *The restriction of $\chi$ to $X_- + \mathfrak{g}_{X_+}$ induces an isomorphism from $X_- + \mathfrak{g}_{X_+}$ onto $\mathfrak{c}$. In particular, it induces an isomorphism from $X_- + \mathfrak{g}_{X_+}(F)$ onto $\mathfrak{c}(F)$.*

[5] *Translator's note.* The source uses $\Delta$, rather than the previously defined $\Phi$, in these two sums. We use $\Phi$ consistently.

We obtain the following corollaries.

**Corollary 4.3.** *Every stable class in $\Sigma(\mathfrak{g})$ contains exactly one element of $X_- + \mathfrak{g}_{X_+}$. The conjugacy class of this element does not depend on the choice of pinning.*

**Corollary 4.4.** *Let $(H, \kappa, \xi)$ be an endoscopic datum for a quasi-split group $G$. Every $G$-regular element of $\mathfrak{h}(F)$ transfers to $\mathfrak{g}$.*

**4.7.** We no longer assume that $G$ is quasi-split. The $\overline{F}$-isomorphism $\varphi$ in (2.1) induces an $F$-isomorphism $\mathfrak{c}_G \to \mathfrak{c}_{G^*}$. The preceding corollary leads us to study the conditions under which $X^* \in \mathfrak{g}^*_{\mathrm{reg}}(F)$ transfers to $\mathfrak{g}$, that is, under which the fiber $\chi_G^{-1}(\chi_{G^*}(X^*))$ contains rational points.

**Lemma 4.5.** *Let $X^* \in \mathfrak{g}^*_{\mathrm{reg}}(F)$. The following conditions are equivalent.*

1. *$X^*$ transfers to $\mathfrak{g}$.*
2. *There exist $g \in G^*_{\mathrm{sc}}$ and $X \in \mathfrak{g}_{\mathrm{reg}}(F)$ such that $\mathrm{Ad}(g) \circ \varphi(X) = X^*$.*
3. *The class of $\varphi$ in $H^1(F, G^*_{\mathrm{ad}})$ belongs to the image of*
$$H^1(F, T_{X^*,\mathrm{ad}}) \longrightarrow H^1(F, G^*_{\mathrm{ad}}).$$

*If $F$ is nonarchimedean local, every elliptic element of $\mathfrak{g}^*(F)$ transfers to $\mathfrak{g}$.*

*Proof.* Clearly, assertions 1 and 2 are equivalent, and 2 implies 3. We therefore only prove that 3 implies 2. To simplify notation, put $T = T_{X^*}$. Let $u_\sigma$ be the cocycle with values in $G^*_{\mathrm{ad}}$ defined by $\varphi\sigma(\varphi)^{-1}$. Assertion 3 says that its class lies in the image of
$$H^1(F, T_{\mathrm{ad}}) \longrightarrow H^1(F, G^*_{\mathrm{ad}}),$$
that is, there is $g \in G^*_{\mathrm{sc}}$ such that $gu_\sigma\sigma(g)^{-1}$ belongs to $T/Z_{G^*}$. Let $X \in \mathfrak{g}$ satisfy $\mathrm{Ad}(g) \circ \varphi(X) = X^*$. One checks that $X \in \mathfrak{g}_{\mathrm{reg}}(F)$.

We now prove the last assertion, assuming that $F$ is nonarchimedean local. Since $X^*$ is elliptic, $T$ is elliptic, and the Tate–Nakayama isomorphism gives $H^2(F, T_{\mathrm{sc}}) = 1$. We must prove that
$$H^1(F, T_{\mathrm{ad}}) \longrightarrow H^1(F, G^*_{\mathrm{ad}})$$
is surjective. The exact sequence
$$H^1(F, T_{\mathrm{ad}}) \longrightarrow H^1(F, T_{\mathrm{sc}} \to T_{\mathrm{ad}}) \longrightarrow H^2(F, T_{\mathrm{sc}})$$
implies the surjectivity of $H^1(F, T_{\mathrm{ad}}) \to H^1(F, T_{\mathrm{sc}} \to T_{\mathrm{ad}})$.[6] The desired surjectivity now follows from the commutative diagram
$$\begin{array}{ccc} H^1(F, T_{\mathrm{ad}}) & \longrightarrow & H^1(F, G^*_{\mathrm{ad}}) \\ \downarrow & & \downarrow \\ H^1(F, T_{\mathrm{sc}} \to T_{\mathrm{ad}}) & \longrightarrow & H^1(F, G^*_{\mathrm{sc}} \to G^*_{\mathrm{ad}}) \end{array}$$
and the injectivity of the abelianization map, the right vertical arrow, over a nonarchimedean local field, since $H^1(F, G^*_{\mathrm{sc}}) = 1$ over such a field. □

[6] *Translator's note.* The target of this map is truncated in the source. In the following diagram the source also suppresses the stars on $G$; they are restored to agree with the map being studied.

**4.8.** The following lemma is an immediate consequence of the preceding corollary and lemma.

**Lemma 4.6.** *Let $F$ be a nonarchimedean local field. Suppose that the endoscopic datum $(H, \kappa, \xi)$ is elliptic. Then every elliptic $G$-regular element of $\mathfrak{h}(F)$ transfers to $G$.*

## 5. Harmonic analysis on $p$-adic Lie algebras

**5.1.** Throughout this section, $F$ is a $p$-adic field. We recall some results about harmonic analysis on $\mathfrak{g}(F)$ that will be useful later. An excellent reference is [2].

**5.2.** Let $C_c^\infty(\mathfrak{g}(F))$ be the vector space of smooth, that is, locally constant, compactly supported complex-valued functions on $\mathfrak{g}(F)$. Let $\mathcal{O}_F$ be the ring of integers of $F$. By a *lattice* in $\mathfrak{g}(F)$, we mean an $\mathcal{O}_F$-submodule that is both open and compact. If $R$ is such a lattice, write $C_c^\infty(\mathfrak{g}(F)/R)$ for the subspace of $C_c^\infty(\mathfrak{g}(F))$ consisting of $R$-invariant functions.

**5.3.** By a *distribution* on $\mathfrak{g}(F)$, we mean a linear form on $C_c^\infty(\mathfrak{g}(F))$. If $T$ is a distribution, its support is denoted by $\mathrm{supp}(T)$. The action of $G(F)$ on $\mathfrak{g}(F)$ defines, by duality, an action on $C_c^\infty(\mathfrak{g}(F))$ and on the space of distributions. A distribution is $G(F)$-*invariant*, or simply *invariant*, if it is fixed by this action. Let $J$ be the space of invariant distributions.

For subsets $\omega \subset \mathfrak{g}(F)$ and $H \subset G(F)$, set

$$\omega^H = \{\mathrm{Ad}(x)X \mid x \in H \text{ and } X \in \omega\}.$$

For $\omega \subset \mathfrak{g}(F)$, let $J(\omega)$ be the space of invariant distributions $T$ whose support is contained in the closure of $\omega^{G(F)}$.

Here is a first fundamental finiteness result, due to Howe and Harish-Chandra; it is Theorem 12.1 of [2].

**Theorem 5.1.** *Let $\omega \subset \mathfrak{g}(F)$ be compact and let $R$ be a lattice in $\mathfrak{g}(F)$. Let $J_R(\omega)$ be the space spanned by the restrictions to $C_c^\infty(\mathfrak{g}(F)/R)$ of the elements of $J(\omega)$. Then*

$$\dim_{\mathbb{C}} J_R(\omega) < \infty.$$

**5.4.** Let

$$\langle\ ,\ \rangle : \mathfrak{g}(F) \times \mathfrak{g}(F) \longrightarrow F$$

be a nondegenerate symmetric bilinear form invariant under $G(F)$.[7] Let $\psi : F \to \mathbb{C}^\times$ be a continuous nontrivial character. Every subspace of $\mathfrak{g}(F)$ on which the restriction of this form is nondegenerate is equipped with the self-dual Haar measure for the bicharacter $\psi(\langle\ ,\ \rangle)$.

We define a Fourier transformation, a linear bijection from $C_c^\infty(\mathfrak{g}(F))$ onto itself, by

$$\widehat{f}(X) = \int_{\mathfrak{g}(F)} f(Y)\psi(\langle X, Y\rangle)\, dY.$$

By duality, we define a Fourier transform on the space of distributions. The action of $G(F)$ commutes with Fourier transformation. In particular, Fourier transformation preserves $J$.

[7] *Translator's note.* The source gives the target as $\mathbb{C}$ here and in §11.2. We write $F$, as required for the subsequent composition with the character $\psi$ of $F$, or of $\mathbb{A}_F/F$.

**5.5.** We can now state Harish-Chandra's regularity theorem; see [2, Theorem 4.4].

**Theorem 5.2.** *Let $\omega \subset \mathfrak{g}(F)$ be compact and let $T \in J(\omega)$. There exists a function $\widehat{t}$, locally constant on $\mathfrak{g}_{\mathrm{reg}}(F)$ and locally integrable on $\mathfrak{g}(F)$, such that*

$$\widehat{T}(f) = \int_{\mathfrak{g}(F)} \widehat{t}(X) f(X)\, dX.$$

## 6. Orbital integrals over a $p$-adic field

**6.1.** We retain the notation of the preceding section.

**6.2.** Fix a Haar measure on $G(F)$. Let $\mathcal{T}^G$ be the set of $G(F)$-conjugacy classes of maximal $F$-subtori of $G$. In what follows, we identify this set with a set of representatives. Equip the tori in $\mathcal{T}^G$ with Haar measures. Every maximal $F$-subtorus of $G$ is conjugate to a torus in $\mathcal{T}^G$; it is then equipped with the Haar measure obtained by transport.

**6.3.** Let $X \in \mathfrak{g}_{\mathrm{reg}}(F)$. The *orbital integral* of $X$ is the invariant distribution $I_X$ defined by

$$I_X(f) = |D^{\mathfrak{g}}(X)|_F^{1/2} \int_{T_X(F)\backslash G(F)} f(\mathrm{Ad}(x^{-1})X)\, dx,$$

where $dx$ is the quotient Haar measure and $D^{\mathfrak{g}}(X) = \det(\mathrm{ad}(X)|_{\mathfrak{g}/\mathfrak{t}_X})$ is the Weyl discriminant.[8] The integral converges. This follows because the function

$$x \in T_X(F)\backslash G(F) \longmapsto f(\mathrm{Ad}(x^{-1})X)$$

has compact support: indeed, $x \mapsto \mathrm{Ad}(x^{-1})X$ induces a homeomorphism from $T_X(F)\backslash G(F)$ onto the orbit of $X$, and this orbit is closed because $X$ is semisimple.

The importance of regular semisimple orbital integrals in harmonic analysis comes from the following theorem of Harish-Chandra: regular semisimple orbital integrals are weakly dense in $J$, in the sense that, if $I_X(f) = 0$ for every $X \in \mathfrak{g}_{\mathrm{reg}}(F)$, then $T(f) = 0$ for every $T \in J$; see [2, Theorem 3.1].

**6.4.** A fundamental problem is to characterize the functions

$$X \in \mathfrak{g}_{\mathrm{reg}}(F) \longmapsto I_X(f)$$

obtained as $f$ ranges over $C_c^\infty(\mathfrak{g}(F))$. In general, this is a very difficult problem. When we restrict to $f \in C_c^\infty(\mathfrak{g}_{\mathrm{reg}}(F))$, that is, smooth functions whose support is a compact subset of $\mathfrak{g}_{\mathrm{reg}}(F)$, the problem is easy.

**Lemma 6.1.** *Let $I : \mathfrak{g}_{\mathrm{reg}}(F) \to \mathbb{C}$. The following conditions are equivalent.*[9]

1. *$I$ is locally constant, invariant under $G(F)$, and has support contained in a set $\omega^{G(F)}$ with $\omega$ compact.*
2. *There is $f \in C_c^\infty(\mathfrak{g}_{\mathrm{reg}}(F))$ such that*

$$I(X) = I_X(f) \qquad \text{for every } X \in \mathfrak{g}_{\mathrm{reg}}(F).$$

[8] *Translator's note.* The source writes ad$(x)$ in the definition of $D^{\mathfrak{g}}(X)$; the capital $X$ is restored here.

[9] *Translator's note.* Condition 1 is translated as printed. The source does not explicitly require $\omega \subset \mathfrak{g}_{\mathrm{reg}}(F)$, although the converse proof below uses compactness within the regular locus. No extra hypothesis has been inserted into the statement.

*Proof.* Let $T \in \mathcal{T}^G$ and consider

$$\Phi_T : T(F)\backslash G(F) \times \mathfrak{t}_{\mathrm{reg}}(F) \longrightarrow \mathfrak{g}_{\mathrm{reg}}(F),$$
$$(g, X) \longmapsto \mathrm{Ad}(g^{-1})X.$$

A little differential calculus shows that $\Phi_T$ induces an analytic covering of an open subset $\mathfrak{g}_T(F) \subset \mathfrak{g}_{\mathrm{reg}}(F)$ by $T(F)\backslash G(F) \times \mathfrak{t}_{\mathrm{reg}}(F)$; this covering is Galois with group $W^G(T)$. The open sets $\mathfrak{g}_T(F)$, for $T \in \mathcal{T}^G$, are pairwise disjoint and their union is $\mathfrak{g}_{\mathrm{reg}}(F)$.

Let $f \in C_c^\infty(\mathfrak{g}_{\mathrm{reg}}(F))$. We show that $X \mapsto I_X(f)$ satisfies condition 1. Only local constancy is not immediate. It is enough to show that, for every $T \in \mathcal{T}^G$, the function $(g, X) \mapsto I_{\Phi_T(g,X)}(f)$ is locally constant.[10] Now $f \circ \Phi_T$ belongs to $C_c^\infty(T(F)\backslash G(F) \times \mathfrak{t}_{\mathrm{reg}}(F))$. Thus there are finite families $\alpha_i \in C_c^\infty(T(F)\backslash G(F))$ and $\theta_i \in C_c^\infty(\mathfrak{t}_{\mathrm{reg}}(F))$ such that

$$f \circ \Phi_T = \sum_i \alpha_i \otimes \theta_i.$$

For $(g, X) \in T(F)\backslash G(F) \times \mathfrak{t}_{\mathrm{reg}}(F)$, we calculate

$$\begin{aligned} I_{\Phi_T(g,X)}(f) &= |D^{\mathfrak{g}}(X)|_F^{1/2} \int_{T_{\mathrm{Ad}(g^{-1})X}(F)\backslash G(F)} f(\mathrm{Ad}(x^{-1}g^{-1})X)\, dx \\ &= |D^{\mathfrak{g}}(X)|_F^{1/2} \int_{T_X(F)\backslash G(F)} f(\mathrm{Ad}(x^{-1})X)\, dx \\ &= |D^{\mathfrak{g}}(X)|_F^{1/2} \sum_i \left( \int_{T_X(F)\backslash G(F)} \alpha_i(x)\, dx \right) \theta_i(X), \end{aligned}$$

and the result is now evident.

Conversely, suppose that $I$ satisfies condition 1. Let $T \in \mathcal{T}^G$. The function $I \circ \Phi_T$ is constant in the first variable and defines a function $\theta \in C_c^\infty(\mathfrak{t}_{\mathrm{reg}}(F))$ of the second variable. Its support is regular and compact because

$$\mathrm{supp}(\theta) \subset \mathfrak{t}_{\mathrm{reg}}(F) \cap \omega^{G(F)},$$

and the latter set is compact. Let $\alpha \in C_c^\infty(T(F)\backslash G(F))$ be $W$-invariant and satisfy[11]

$$\int_{T_X(F)\backslash G(F)} \alpha(x)\, dx = 1.$$

Then $\alpha \otimes |D^{\mathfrak{g}}(\cdot)|_F^{-1/2}\theta$ is $W$-invariant and belongs to $C_c^\infty(T(F)\backslash G(F) \times \mathfrak{t}_{\mathrm{reg}}(F))$. It therefore defines $f_T \in C_c^\infty(\mathfrak{g}_T(F))$. Letting $T$ vary, we obtain a function $f$ satisfying condition 2. □

The following lemma is then immediate.

**Lemma 6.2.** *Let $f \in C_c^\infty(\mathfrak{g}_{\mathrm{reg}}(F))$. The function $X \mapsto I_X(f)$ satisfies the assertions of the preceding lemma.*

[10] *Translator's note.* The corresponding sentence in the source breaks off after the notation $I_{\Phi_T}(f)$. The asserted local constancy is made explicit here.

[11] *Translator's note.* In this converse argument the source says "first variable" where $\theta$ is a function of the second, and writes $\alpha_i$ instead of the newly chosen $\alpha$ in the normalization. These two slips are corrected.

**6.5.** Here is a more precise and slightly different formulation of Harish-Chandra's regularity theorem, Theorem 5.2, for orbital integrals.

**Theorem 6.3.** *There is a unique locally constant $G(F) \times G(F)$-invariant function*

$$\widehat{i}^G : \mathfrak{g}_{\mathrm{reg}}(F) \times \mathfrak{g}_{\mathrm{reg}}(F) \longrightarrow \mathbb{C}$$

*such that, for every $X \in \mathfrak{g}_{\mathrm{reg}}(F)$ and $f \in C_c^\infty(\mathfrak{g}(F))$,*

$$\widehat{I}_X(f) = \sum_{T \in \mathcal{T}^G} |W^G(T)|^{-1} \int_{\mathfrak{t}(F)} I_Y(f) \widehat{i}^G(X, Y)\, dY.$$

The two formulations are related by the Weyl integration formula: there are constants $c_T$, for $T \in \mathcal{T}^G$, such that, for every $f \in C_c^\infty(\mathfrak{g}(F))$,

$$\int_{\mathfrak{g}(F)} f(X)\, dX = \sum_{T \in \mathcal{T}^G} c_T |W^G(T)|^{-1} \int_{\mathfrak{t}(F)} |D^{\mathfrak{g}}(X)|_F \cdot \int_{T(F)\backslash G(F)} f(\mathrm{Ad}(x^{-1})X)\, dx\, dX.$$

Local constancy of $\widehat{i}^G$ in the first variable follows from Lemma 6.2.

*Remark.* It is clear that $\widehat{i}^G$ does not depend on the choice of Haar measures on $G(F)$ and the tori $T(F)$.

## 7. Transfer factors

**7.1.** In this section, $F$ is a local field.

**7.2.** Let $(H, \kappa, \xi)$ be an endoscopic datum for $G$.[12] The purpose of this section is to define the Langlands–Shelstad *transfer factor*. It is a map

$$\Delta : \Sigma_G(\mathfrak{h}) \times \Gamma(\mathfrak{g}) \longrightarrow \mathbb{C}$$

such that $\Delta(Y, X) \neq 0$ if and only if $Y \to X$. We now specify its nonzero values. To do so, we define a canonical relative transfer factor $\Delta(Y_1, X_1, Y_2, X_2)$ for elements $Y_i \in \Sigma_G(\mathfrak{h})$ and $X_i \in \Gamma(\mathfrak{g})$ satisfying $Y_i \to X_i$ for $i = 1, 2$. Fix a base pair $(Y_0, X_0)$ such that $Y_0 \to X_0$, provided such a pair exists. Choose $\Delta(Y_0, X_0) \in \mathbb{C}^\times$ arbitrarily. For every pair $(Y, X)$ such that $Y \to X$, set

$$\Delta(Y, X) = \Delta(Y, X, Y_0, X_0)\Delta(Y_0, X_0).$$

This makes sense because, as we shall see, for such a pair one has

$$\Delta(Y, X, Y, X) = 1. \tag{7.1}$$

To define the relative transfer factor, we need a quasi-split inner form $G^*$ of $G$, as in §2.5 and (2.1), together with a pinning of $G^*$, as in §4.6. We then have a section of $\chi_{G^*}$ by Theorem 4.2. Composing it with $\mathfrak{c}_H \to \mathfrak{c}_{G^*}$ from (4.1) and with $\chi_H$ gives a map

$$\Sigma_G(\mathfrak{h}) \longrightarrow X_- + \mathfrak{g}_{X_+}(F),$$

which we denote by $Y \mapsto Y^*$. We leave it to the reader to check that the relative transfer factor is independent of these auxiliary choices.

[12] *Translator's note.* The source says "an endoscopic datum for $H$" here; the ambient group is $G$.

**7.3.** Underlying the datum $(H, \kappa, \xi)$ are maximal subtori $T_G \subset G$ and $T_H \subset H$. Moreover, $\kappa$ induces a character of $H^1(F, T_{H_{\mathrm{sc}}} \to T_{G_{\mathrm{sc}}})$; see §3.1. Set

$$U = (T_{G_{\mathrm{sc}}} \times T_{G_{\mathrm{sc}}})/\{(z, z^{-1}) \mid z \in Z_{G_{\mathrm{sc}}}\}.$$

Multiplication induces an evident morphism $U \to T_{G_{\mathrm{sc}}}$ and, more generally, a morphism of complexes

$$[T_{H_{\mathrm{sc}}} \times T_{H_{\mathrm{sc}}} \to U] \longrightarrow [T_{H_{\mathrm{sc}}} \to T_{G_{\mathrm{sc}}}].$$

Thus we have a map

$$H^1(F, T_{H_{\mathrm{sc}}} \times T_{H_{\mathrm{sc}}} \to U) \longrightarrow H^1(F, T_{H_{\mathrm{sc}}} \to T_{G_{\mathrm{sc}}}).$$

Let $\kappa_U$ be the character that $\kappa$ induces on the first group.

We wish to define $\Delta(Y_1, X_1, Y_2, X_2)$ for a quadruple as above. By the definition of $Y_i^*$, there is $g_i \in G_{\mathrm{sc}}^*$ such that $\mathrm{Ad}(g_i)T_G = T_{Y_i^*}$ and the conditions of §4.4 hold. Set

$$U_{1,2} = (T_{Y_1^*, G_{\mathrm{sc}}} \times T_{Y_2^*, G_{\mathrm{sc}}})/\{(z, z^{-1}) \mid z \in Z_{G_{\mathrm{sc}}}\}.$$

By a variant of the argument following (3.3), the pair $(\mathrm{Ad}(g_1^{-1}), \mathrm{Ad}(g_2^{-1}))$ induces a $\Gamma$-equivariant isomorphism

$$\begin{aligned} &X_*(T_{Y_1, H_{\mathrm{sc}}} \times T_{Y_2, H_{\mathrm{sc}}}) \backslash X_*(U_{1,2}) \\ &\longrightarrow X_*(T_{H_{\mathrm{sc}}} \times T_{H_{\mathrm{sc}}}) \backslash X_*(U), \end{aligned}$$

and hence an isomorphism[13]

$$\begin{aligned} &H^1(F, T_{Y_1, H_{\mathrm{sc}}} \times T_{Y_2, H_{\mathrm{sc}}} \to U_{1,2}) \\ &\longrightarrow H^1(F, T_{H_{\mathrm{sc}}} \times T_{H_{\mathrm{sc}}} \to U). \end{aligned}$$

Therefore $\kappa_U$ induces a character on $H^1(F, T_{Y_1, H_{\mathrm{sc}}} \times T_{Y_2, H_{\mathrm{sc}}} \to U_{1,2})$, and hence a character $\kappa_{1,2}$ on $H^1(F, U_{1,2})$.

Since $Y_i \to X_i$ by assumption, we have $\chi_G(X_i) = \chi_{G^*}(Y_i^*)$. Thus there is $x_i \in G_{\mathrm{sc}}^*$ such that

$$\mathrm{Ad}(x_i) \circ \varphi(X_i) = Y_i^*.$$

For every $\sigma \in \Gamma$, let $u_\sigma \in G_{\mathrm{sc}}^*$ satisfy $\varphi\sigma(\varphi)^{-1} = \mathrm{Ad}(u_\sigma)$. Then

$$\left(x_1 u_\sigma \sigma(x_1)^{-1},\ \sigma(x_2) u_\sigma^{-1} x_2^{-1}\right)$$

defines a 1-cochain with values in $U_{1,2}$, whose class in $H^1(F, U_{1,2})$ is denoted by $\mathrm{inv}(Y_1, X_1, Y_2, X_2)$. Set

$$\Delta(Y_1, X_1, Y_2, X_2) = \langle \kappa_{1,2}, \mathrm{inv}(Y_1, X_1, Y_2, X_2) \rangle.$$

[13] *Translator's note.* The source repeats $T_{Y, H_{\mathrm{sc}}}$ in both factors in these formulas and the next sentence. The subscripts $Y_1$ and $Y_2$ are restored. The source's suppression of stars on the tori of the chosen inner form is otherwise retained.

**7.4.** Let $X^* \in \mathfrak{g}^*_{\mathrm{reg}}(F)$. For $i = 1, 2$, let $X_i \in \mathfrak{g}(F)$ be such that there exists $x_i \in G^*_{\mathrm{sc}}$ with $\mathrm{Ad}(x_i) \circ \varphi(X_i) = X^*$. Then the product

$$x_1 u_\sigma \sigma(x_1)^{-1} \sigma(x_2) u_\sigma^{-1} x_2^{-1}$$

defines a Galois 1-cochain with values in $T_{X^*,\mathrm{sc}}$. Its class in $H^1(F, T_{X^*,\mathrm{sc}})$ is denoted by $\mathrm{inv}(X^*; X_1, X_2)$. One easily checks the following formula: for every $Y \in \mathfrak{h}_{G\text{-reg}}(F)$ such that $Y \to X^*$,

$$\Delta(Y, X_1) = \Delta(Y, X_2)\langle \kappa_Y, \mathrm{inv}(X^*; X_1, X_2)\rangle, \tag{7.2}$$

where $\kappa_Y$ was defined at the end of §4.4.

## 8. Smooth transfer and Fourier transformation: the $p$-adic case

**8.1.** Throughout this section, $F$ is a $p$-adic field.

**8.2.** Two stably conjugate elements of $\mathfrak{g}_{\mathrm{reg}}(F)$ have "stably conjugate," hence isomorphic, centralizer tori. We require the Haar measures on these tori to correspond under this isomorphism.

**8.3.** Let $X \in \Sigma(\mathfrak{g})$. The *stable orbital integral* $S_X$ is the invariant distribution defined by

$$S_X(f) = \sum_{\substack{Z \in \Gamma(\mathfrak{g}) \\ \chi(Z) = \chi(X)}} I_Z(f).$$

A distribution $T$ is *stable* if it belongs to the weak closure of the space of stable orbital integrals, that is, if $S_X(f) = 0$ for every $X \in \Sigma(\mathfrak{g})$ implies $T(f) = 0$.

To emphasize that these objects are attached to $G$, we will write $I^G$, $S^G$, and so on.

**8.4.** Let $(H, \kappa, \xi)$ be an endoscopic datum for $G$. For every $Y \in \Sigma_G(\mathfrak{h})$, the *endoscopic orbital integral* $I_Y^{G,H}$ is the invariant distribution defined by

$$I_Y^{G,H}(f) = \sum_{X \in \Gamma(\mathfrak{g})} \Delta(Y, X) I_X(f).$$

**8.5.** Here is Waldspurger's smooth transfer theorem.

**Theorem 8.1.** *For every $f \in C_c^\infty(\mathfrak{g}(F))$, there exists $f^H \in C_c^\infty(\mathfrak{h}(F))$ such that, for every $Y \in \Sigma_G(\mathfrak{h})$,*

$$I_Y^{G,H}(f) = S_Y^H(f^H).$$

*We say that $f^H$ is a* transfer *of $f$.*

One can show that, if $T$ is a stable distribution on $\mathfrak{h}(F)$, then $T(f^H)$ depends only on $f$. This gives a distribution on $\mathfrak{g}(F)$, the *dual endoscopic transfer*.

**8.6.** Suppose that $G$ is unramified: by definition, it is quasi-split and splits over an unramified extension. Bruhat–Tits theory shows that there is a smooth group scheme $\mathcal{G}$ over the ring of integers $\mathcal{O}_F$, with connected reductive fibers and generic fiber $G$. Let $1_G$ be the characteristic function of the lattice $\mathrm{Lie}(\mathcal{G})(\mathcal{O}_F)$ in $\mathfrak{g}(F)$.[14] Such a lattice is called *hyperspecial*. If $H$ is unramified, we likewise have a function $1_H$. The following

[14] *Translator's note.* The source says "special fiber" instead of "generic fiber," and writes $\mathcal{G}(\mathcal{O}_F)$ for a lattice in the Lie algebra. The generic fiber and Lie algebra lattice are made explicit here.

theorem, due to Waldspurger for $\mathrm{SL}_n$, to Laumon–Ngô for unitary groups, and to Ngô in general, holds for every choice of integral models as above: it is the Langlands–Shelstad fundamental lemma.

**Theorem 8.2.** *There is a constant $c \neq 0$ such that, for every $Y \in \Sigma_G(\mathfrak{h})$,*

$$c\, I_Y^{G,H}(1_G) = S_Y^H(1_H).$$

**8.7.** Harish-Chandra emphasized the analogy between harmonic analysis on the group $G$ and on its Lie algebra $\mathfrak{g}$. On both sides, of course, there are orbital integrals attached to regular semisimple elements. There are, however, other distributions on the group, namely the traces of tempered representations. Their analogues on the Lie algebra are the Fourier transforms of orbital integrals.

Waldspurger carried this analogy further. On the group, tempered representations are organized into $L$-packets whose associated traces should be stable distributions. According to Waldspurger, the analogue of an $L$-packet is a stable conjugacy class. In particular, one expects the Fourier transform $\widehat{S}_X^G$, for $X \in \Sigma(\mathfrak{g})$, to be stable. Waldspurger proved the following theorem.

**Theorem 8.3.** *The Fourier transform of a stable distribution is stable.*

**8.8.** As the bilinear form on $\mathfrak{h}(F)$, as in §5.4, choose the form whose restriction to $\mathfrak{t}_H$ is $\langle \iota(\cdot), \iota(\cdot)\rangle$; see §3.1. Using the same character $\psi$ gives a Fourier transformation. We can then consider the dual transfer of $\widehat{S}_Y^H$. What is it? Waldspurger's answer is the following.

**Theorem 8.4.** *There is a constant $c \neq 0$ such that, for every $Y \in \Sigma_G(\mathfrak{h})$ and every $f \in C_c^\infty(\mathfrak{g}(F))$,*

$$c\, \widehat{I}_Y^{G,H}(f) = \widehat{S}_Y^H(f^H),$$

*where $f^H$ is a transfer of $f$. In other words, transfer commutes with Fourier transformation up to a constant.*

**8.9.** Theorem 8.4 is equivalent to the following theorem, also due to Waldspurger. The forward implication follows from the local constancy of the functions $\widehat{i}^G$ and from Lemma 6.1, which allows one to isolate orbits.

**Theorem 8.5.** *There is a constant $c \neq 0$ such that, for every $Y \in \Sigma_G(\mathfrak{h})$ and every $Z \in \Gamma(\mathfrak{g})$, the expression*

$$\sum_{X \in \Gamma(\mathfrak{g})} \Delta(Y, X) \widehat{i}^G(X, Z)$$

*is equal to*

$$c \sum_{Z' \in \Gamma(\mathfrak{h})} \sum_{Y'} \frac{\widehat{i}^H(Y', Z')\Delta(Z', Z)}{|\ker(H^1(F, T_{Z'}) \to H^1(F, H))|}, \tag{8.1}$$

*where the sum over $Y'$ runs over the classes $Y' \in \Gamma(\mathfrak{h})$ stably conjugate to $Y$.*[15]

[15] *Translator's note.* The source writes $Z \in \Gamma(X)$ and uses $\widehat{i}^H(Y, Z')$ in the summand despite summing over $Y'$. We have written $\Gamma(\mathfrak{g})$ and $\widehat{i}^H(Y', Z')$, respectively. The scalar is left with the original notation $c$; the equivalence of the formulations does not identify this scalar with the identically named scalar in Theorem 8.4.

**8.10.** Remarkably, Theorem 8.5 implies the existence of transfer in Theorem 8.1. Let us sketch why. Fix a compact subset $\omega$ of $\mathfrak{h}$. For every compact $K \subset \mathfrak{g}(F)$, Howe's finiteness theorem, Theorem 5.1, implies that the space $V_{\omega,K}$ spanned by the functions

$$Y \in \omega \cap \mathfrak{h}_{G\text{-reg}}(F) \longmapsto \sum_{X \in \Gamma(\mathfrak{g})} \Delta(Y,X)\widehat{i}^G(X,Z), \qquad Z \in K \cap \mathfrak{g}_{\mathrm{reg}}(F),$$

is finite-dimensional. By Theorem 8.5, this space is contained in the span of the functions in (8.1), and the latter is contained in the space of functions

$$Y \in \omega \cap \mathfrak{h}_{G\text{-reg}}(F) \longmapsto S_Y^H(f'), \qquad f' \in C_c^\infty(\mathfrak{h}(F));$$

to see this, use Lemma 6.1.

Take $f \in C_c^\infty(\mathfrak{g}(F))$. The integration formula in Theorem 6.3 implies that the function

$$Y \in \omega \cap \mathfrak{h}_{G\text{-reg}}(F) \longmapsto I_Y^{G,H}(f)$$

belongs to $V_{\omega,K}$ for a suitable $K$. But, as we have said, every element of $V_{\omega,K}$ has the form $S_Y^H(f')$ for some $f' \in C_c^\infty(\mathfrak{h}(F))$.

This gives local transfer near every point. A partition of unity allows us to glue the resulting transfers.

**8.11.** The use of an exponential map and descent to centralizers shows that transfer for groups follows from transfer for the Lie algebras of the centralizers of semisimple elements of the group. This requires checking various descent formulas for transfer factors.

## 9. Orbital integrals on complex Lie algebras

**9.1.** In this section, the field is $\mathbb{C}$. Over this field, stable conjugacy is the same as conjugacy. Let $(H,\kappa,\xi)$ be an endoscopic datum for $G$. The transfer factor is either a fixed nonzero constant or zero.

**9.2.** Let $B \subset G$ be a Borel subgroup, $T \subset B$ a maximal subtorus, and $K \subset G(\mathbb{C})$ a maximal compact subgroup in "good position" with respect to $T$. Let $U_B$ be the unipotent radical of $B$. Let $\mathcal{S}(\mathfrak{g}(\mathbb{C}))$ be the Schwartz space of $\mathfrak{g}(\mathbb{C})$. For $X \in \Gamma(\mathfrak{g})$ and $f \in \mathcal{S}(\mathfrak{g}(\mathbb{C}))$, define the orbital integral $I_X(f)$ as in §6.3. This integral converges. In the complex case, the Iwasawa decomposition expresses an orbital integral as a "spherical" constant term. More precisely, with choices of Haar measures that we do not specify here, for every $X \in \mathfrak{t}_{\mathrm{reg}}(\mathbb{C})$ one has

$$I_X(f) = f_B(X),$$

where the "constant term" is defined, for every $X \in \mathfrak{t}(\mathbb{C})$, by[16]

$$f_B(X) = \int_{K \times \mathfrak{u}_B(\mathbb{C})} f\Big(\mathrm{Ad}(k)(X+N)\Big)\, dN\, dk.$$

We have the following theorem, a special case of a theorem of Bouaziz; see [1].

[16] *Translator's note.* The source writes $K \times U_B(\mathbb{C})$, although the integration variable $N$ is added to $X$ in the Lie algebra. We write $\mathfrak{u}_B = \mathrm{Lie}(U_B)$ here.

**Theorem 9.1.** *Let $W = W^G(T)$ be the Weyl group. The map*

$$f \in \mathcal{S}(\mathfrak{g}(\mathbb{C})) \longmapsto f_B \in \mathcal{S}(\mathfrak{t}(\mathbb{C}))^W$$

*is surjective.*[17]

**9.3.** Fix a character $\psi$ and a form as in §5.4, giving a Fourier transformation on $\mathcal{S}(\mathfrak{g}(\mathbb{C}))$. We require the forms on $\mathfrak{g}$ and $\mathfrak{h}$ to be compatible, as in §8.8: they agree on the Lie algebra of "the" common torus, where we say "the" because all maximal tori are conjugate.[18] The following result is then easily deduced from the preceding theorem.

**Theorem 9.2.** *Let $f \in \mathcal{S}(\mathfrak{g}(\mathbb{C}))$. Then*

– *there is a transfer $f^H \in \mathcal{S}(\mathfrak{h}(\mathbb{C}))$ such that, for every $Y \in \Gamma_G(\mathfrak{h})$,*

$$I_Y(f^H) = I_Y^{G,H}(f);$$

– *transfer commutes with Fourier transformation.*

## 10. Pre-stabilization of the trace formula for Lie algebras

**10.1.** In this section, unless otherwise stated, $F$ is a number field.

**10.2. An analogue of the elliptic part of the Arthur–Selberg trace formula.** Let $\mathcal{S} = \mathcal{S}(\mathfrak{g}(\mathbb{A}))$ be the Schwartz–Bruhat space of $\mathfrak{g}(\mathbb{A})$: it is the space spanned by functions $f_\infty \otimes f^\infty$, where $f_\infty$ is a Schwartz function on $\mathfrak{g}(F \otimes \mathbb{R})$ and $f^\infty$ is the characteristic function of a compact open subgroup of $\mathfrak{g}(\mathbb{A}_f)$; here $\mathbb{A}_f$ denotes the finite adeles.[19] Write $G(\mathbb{A})^1$ for the subgroup of $G(\mathbb{A})$ fitting into the exact sequence

$$1 \longrightarrow G(\mathbb{A})^1 \longrightarrow G(\mathbb{A}) \longrightarrow \mathrm{Hom}(X^*(G), \mathbb{R}_+^\times) \longrightarrow 1,$$

where $X^*(G)$ is the group of rational characters of $G$ defined over $F$, and the nonobvious map sends $g \in G(\mathbb{A})$ to the homomorphism $\chi \mapsto |\chi(g)|_{\mathbb{A}}$.

Equip $G(\mathbb{A})$ with Tamagawa measure and $\mathrm{Hom}(X^*(G), \mathbb{R}_+^\times)$ with the measure induced by the usual measure on $\mathbb{R}_+^\times$. This gives a measure on $G(\mathbb{A})^1$. The quotient $G(F)\backslash G(\mathbb{A})^1$ has finite volume, and the *Tamagawa number* of $G$ is[20]

$$\tau(G) = \mathrm{vol}(G(F)\backslash G(\mathbb{A})^1).$$

The calculation of Tamagawa numbers was completed with Kottwitz's proof of Weil's conjecture. We state the result in the following form; see [5, Corollary 1.7.4, p. 39]:[21]

$$\tau(G) = \frac{|H^1_{\mathrm{ab}}(\mathbb{A}/F, G)|}{|\mathrm{Ker}^1_{\mathrm{ab}}(F, G)|}.$$

This expression also applies to subtori of $G$.

[17] *Translator's note.* The source gives the target as $\mathcal{S}(\mathfrak{g}(\mathbb{C}))^W$. It is written here as $\mathcal{S}(\mathfrak{t}(\mathbb{C}))^W$, since $f_B$ was just defined on $\mathfrak{t}(\mathbb{C})$.

[18] *Translator's note.* The source refers to §8.7 here; the compatibility of the forms is specified in §8.8.

[19] *Translator's note.* The source says "subgroup," and that wording is retained. Characteristic functions of compact open cosets are needed to describe the full Schwartz–Bruhat space in this way.

[20] *Translator's note.* The source calls the quotient "finite"; finite volume is meant in the displayed definition.

[21] *Translator's note.* Cardinality bars, absent around the denominator in the source, are supplied here.

**Theorem 10.1.** *For every $f \in \mathcal{S}$, the integral*

$$\int_{G(F)\backslash G(\mathbb{A})^1} \sum_{X \in \mathfrak{g}_{\mathrm{ell}}(F)} |f(x^{-1}Xx)|\, dx < \infty$$

*converges.*

For the proof, we refer to [9, §10.8]. It rests on reduction theory, which provides a substitute for a fundamental domain for $G(F)\backslash G(\mathbb{A})^1$.

We may therefore set

$$J^G(f) = \int_{G(F)\backslash G(\mathbb{A})^1} \sum_{X \in \mathfrak{g}_{\mathrm{ell}}(F)} f(\mathrm{Ad}(x^{-1})X)\, dx.$$

We also write

$$J^G(f) = \sum_{X \in \Gamma_{\mathrm{ell}}(\mathfrak{g})} \tau(T_X) I_X^G(f),$$

where the global orbital integral is defined by

$$I_X^G(f) = \int_{T_X(\mathbb{A})\backslash G(\mathbb{A})} f(\mathrm{Ad}(x^{-1})X)\, dx. \tag{10.1}$$

**10.3.** In the preceding paragraph, we expressed $J^G(f)$ in terms of rational conjugacy classes. We would like an expression in terms of local stable conjugacy classes. This will be possible, but with an error term. We must first introduce the Langlands obstruction.

**10.4. The Langlands obstruction.** We seek a cohomological condition ensuring that an element of $\mathfrak{g}^*(F)$ which transfers to $\mathfrak{g}$ everywhere locally also transfers globally.

More precisely, let $X^* \in \mathfrak{g}^*(F)$ be regular semisimple, and suppose that there exist $x \in G^*(\overline{\mathbb{A}})$ and $X \in \mathfrak{g}(\mathbb{A})$ such that

$$\mathrm{Ad}(x) \circ \varphi(X) = X^*.$$

We may, and do, assume that $x \in G^*_{\mathrm{sc}}(\overline{\mathbb{A}})$; see [9, §10.2]. Let $u_\sigma$ be a cochain with values in $G^*_{\mathrm{sc}}$ such that $\varphi\sigma(\varphi)^{-1} = \mathrm{Ad}(u_\sigma)$ for every $\sigma \in \Gamma$. Let $T$ be the centralizer of $X^*$ in $G^*$. The cochain $xu_\sigma\sigma(x)^{-1} \in T_{\mathrm{sc}}(\overline{\mathbb{A}})$ defines a 1-cocycle class in $H^1(\mathbb{A}_F/F, T_{\mathrm{sc}})$, denoted by $\widetilde{\mathrm{inv}}(X^*, X)$. Since $x$ is well defined up to left translation by an element of $T_{\mathrm{sc}}(\overline{\mathbb{A}})$, the class $\widetilde{\mathrm{inv}}(X^*, X)$ depends only on $X^*$ and $X$. Write $\mathrm{inv}(X^*, X)$ for its image in $\mathcal{E}(T, G, \mathbb{A}/F)$.[22]

The following theorem is due to Langlands.

**Theorem 10.2.** *There exists $g \in G(\mathbb{A})$ such that $\mathrm{Ad}(g)X \in \mathfrak{g}(F)$ if and only if $\mathrm{inv}(X^*, X) = 1$.*

*Proof.* The condition is necessary. Indeed, if $g \in G(\mathbb{A})$ satisfies $\mathrm{Ad}(g)X \in \mathfrak{g}(F)$, we are immediately reduced to the case $X \in \mathfrak{g}(F)$. The set of $y \in G^*_{\mathrm{sc}}$ such that

$$\mathrm{Ad}(y) \circ \varphi(X) = X^*$$

[22] *Translator's note.* The source first calls the lifted class inv, then uses $\widetilde{\mathrm{inv}}$ to distinguish it from its image in $\mathcal{E}$. The tilde is used consistently here.

is then a nonempty subvariety of $G^*_{\mathrm{sc}}$, since it has an $\overline{\mathbb{A}}$-point. Thus we may take $x \in G^*_{\mathrm{sc}}(\overline{F})$.[23] We see that $\widetilde{\mathrm{inv}}(X^*, X) = 1$, proving the first implication.

The converse is more involved. Since $\mathrm{inv}(X^*, X) = 1$, there exist a 1-cocycle $\beta_\sigma$ with values in $T_{\mathrm{sc}}(\overline{\mathbb{A}})$, an element $y \in T(\overline{\mathbb{A}})$, and a 1-cochain $t_\sigma$ with values in $T_{\mathrm{sc}}(\overline{F})$ such that

$$\beta_\sigma = y^{-1}\sigma(y)$$

—here we do not distinguish $\beta_\sigma$ from its image in $T(\overline{\mathbb{A}})$—and

$$xu_\sigma\sigma(x)^{-1} = \beta_\sigma t_\sigma$$

for every $\sigma \in \Gamma$. Since the coboundary $\partial\beta$ is trivial, $\partial u = \partial t$. It follows that the cochain $t_\sigma u_\sigma^{-1}$ is a cocycle "with values in $G_{\mathrm{sc}}$," a somewhat abusive expression meaning that $\varphi^{-1}(t_\sigma u_\sigma^{-1})$ is a cocycle with values in $G_{\mathrm{sc}}$.

The cochain $yxu_\sigma\sigma(yx)^{-1}u_\sigma^{-1}$ defines a locally trivial 1-cocycle with values in $G$; its class therefore belongs to $\mathrm{Ker}^1(F, G)$. Its image in the abelianization $\mathrm{Ker}^1_{\mathrm{ab}}(F, G)$ is trivial: it is the class of the hypercochain $(1, t_\sigma u_\sigma^{-1})$, which is cohomologous to the trivial class. But the abelianization map

$$\mathrm{Ker}^1(F, G) \longrightarrow \mathrm{Ker}^1_{\mathrm{ab}}(F, G)$$

is an isomorphism.[24] Hence the cochain $yxu_\sigma\sigma(yx)^{-1}u_\sigma^{-1}$ is trivial: there is $a \in G^*$ such that

$$yxu_\sigma\sigma(yx)^{-1}u_\sigma^{-1} = au_\sigma\sigma(a)^{-1}u_\sigma^{-1}.$$

Let $Y \in \mathfrak{g}(\overline{F})$ satisfy $\mathrm{Ad}(a) \circ \varphi(Y) = X^*$. Then $Y = \mathrm{Ad}(b)X$, where $b = \varphi^{-1}(a^{-1}yx)$, and one checks that $b \in G(\mathbb{A})$. Consequently,

$$Y \in \mathfrak{g}(\overline{F}) \cap \mathfrak{g}(\mathbb{A}) = \mathfrak{g}(F),$$

as required. □

**10.5. Pre-stabilization.** To express $J^G(f)$ in terms of local stable conjugacy classes, we use Fourier inversion on the group $\mathcal{E}(T_{X^*}, G^*, \mathbb{A}/F)$.[25]

We identify $\Gamma_{\mathrm{ell}}(\mathfrak{g})$ and $\Sigma_{\mathrm{ell}}(\mathfrak{g}^*)$ with sets of representatives. For every $X^* \in \Sigma_{\mathrm{ell}}(\mathfrak{g}^*)$, set

$$\Gamma_{X^*}(\mathfrak{g}) = \{X \in \Gamma_{\mathrm{ell}}(\mathfrak{g}) \mid \exists g \in G^*_{\mathrm{sc}},\ \mathrm{Ad}(g) \circ \varphi(X) = X^*\}.$$

This set may be empty. If it is not, its elements represent the ordinary conjugacy classes in a single stable class of $\mathfrak{g}(F)$ determined by $X^*$. We have the disjoint union

$$\Gamma_{\mathrm{ell}}(\mathfrak{g}) = \coprod_{X^* \in \Sigma_{\mathrm{ell}}(\mathfrak{g}^*)} \Gamma_{X^*}(\mathfrak{g}).$$

[23] *Translator's note.* In this part of the proof the source alternates between $G$, $G_{\mathrm{sc}}$, and the quasi-split inner form. The stars and simply connected subscripts in the preceding two sentences are made consistent with the equation $\mathrm{Ad}(y) \circ \varphi(X) = X^*$.

[24] *Translator's note.* The source repeats the subscript ab on both sides of this map. It is removed from the source of the map here.

[25] *Translator's note.* At this point the source writes $\mathcal{E}(T_{X^*}, \mathfrak{g}^*, f)$. The notation is restored from the Fourier-inversion formula that follows.

Introduce also

$$\Gamma_{X^*}(\mathfrak{g}(\mathbb{A})) = \{X \in \mathfrak{g}(\mathbb{A}) \mid \exists g \in G^*_{\mathrm{sc}}(\overline{\mathbb{A}}), \\ \mathrm{Ad}(g) \circ \varphi(X) = X^*\}/G(\mathbb{A}).$$

Theorem 10.2 implies that the evident map

$$\Gamma_{X^*}(\mathfrak{g}) \longrightarrow \{X \in \Gamma_{X^*}(\mathfrak{g}(\mathbb{A})) \mid \mathrm{inv}(X^*, X) = 1\} \tag{10.2}$$

is surjective. It is not injective. Let $X \in \Gamma_{X^*}(\mathfrak{g})$. Then $\Gamma_{X^*}(\mathfrak{g})$ is naturally in bijection with $\mathcal{D}(T_X, G, F)$. An element of $\Gamma_{X^*}(\mathfrak{g})$ maps to the class of $X$ under (10.2) if and only if it corresponds to an element of

$$\mathrm{Ker}\Big(\mathcal{D}(T_X, G, F) \longrightarrow \mathcal{D}(T_X, G, \mathbb{A})\Big) \\ = \mathrm{Ker}\Big(\mathrm{Ker}^1(F, T_X) \longrightarrow \mathrm{Ker}^1(F, G)\Big).$$

Using the natural bijection between $\mathrm{Ker}^1(F, G)$ and $\mathrm{Ker}^1_{\mathrm{ab}}(F, G)$, and the fact that abelianized cohomology is unaffected by inner twisting, we see that all fibers of (10.2) have the same cardinality, namely

$$d(X^*) = \Big|\mathrm{Ker}\Big(\mathrm{Ker}^1(F, T_{X^*}) \longrightarrow \mathrm{Ker}^1(F, G^*)\Big)\Big|.$$

The expression (10.1) defining $I^G_X(f)$ also applies to every $X \in \mathfrak{g}(\mathbb{A})$. As a function of $X$, the orbital integral is $G(\mathbb{A})$-invariant. Finally, if $X \in \Gamma_{X^*}(\mathfrak{g})$, one checks that $\tau(T_X) = \tau(T_{X^*})$.

We may therefore write

$$J^G(f) = \sum_{X^* \in \Sigma_{\mathrm{ell}}(\mathfrak{g}^*)} \tau(T_{X^*}) d(X^*) \sum_{\substack{X \in \Gamma_{X^*}(\mathfrak{g}(\mathbb{A})) \\ \mathrm{inv}(X^*, X) = 1}} I^G_X(f) \\ = \sum_{X^* \in \Sigma_{\mathrm{ell}}(\mathfrak{g}^*)} \frac{\tau(T_{X^*}) d(X^*)}{|\mathcal{E}(T_{X^*}, G^*, \mathbb{A}/F)|} \\ \cdot \sum_{X \in \Gamma_{X^*}(\mathfrak{g}(\mathbb{A}))} \sum_{\kappa \in \mathcal{K}(T_{X^*}, G^*, F)} \langle \kappa, \mathrm{inv}(X^*, X)\rangle I^G_X(f).$$

We now use the following exact sequence from [5, Proposition 1.8.4, p. 42]:

$$1 \longrightarrow \mathrm{Ker}\Big(\mathrm{Ker}^1(F, T_{X^*}) \to \mathrm{Ker}^1_{\mathrm{ab}}(F, G^*)\Big) \\ \longrightarrow \mathrm{Ker}^1(F, T_{X^*}) \longrightarrow \mathrm{Ker}^1_{\mathrm{ab}}(F, G^*) \\ \longrightarrow \mathcal{E}(T_{X^*}, G^*, \mathbb{A}/F) \longrightarrow H^1(\mathbb{A}/F, T_{X^*}) \\ \longrightarrow H^1_{\mathrm{ab}}(\mathbb{A}/F, G^*) \longrightarrow H^2(\mathbb{A}/F, T_{X^*,\mathrm{sc}}) \longrightarrow 1.$$

Since $T_{X^*,\mathrm{sc}}$ is elliptic, the Tate–Nakayama isomorphism shows that $H^2(\mathbb{A}/F, T_{X^*,\mathrm{sc}}) = 1$. Using the natural bijection between $\mathrm{Ker}^1(F, G^*)$ and its abelianization, we obtain

$$\frac{\tau(T_{X^*}) d(X^*)}{|\mathcal{E}(T_{X^*}, G^*, \mathbb{A}/F)|} = \tau(G^*).$$

The set $\Gamma_{X^*}(\mathfrak{g}(\mathbb{A}))$ is infinite. There is a finite set $S$ of places such that $G$ is defined over $\mathfrak{o}_S$, the ring of elements of $F$ integral outside $S$, and such that, for all $X, X' \in \Gamma_{X^*}(\mathfrak{g}(\mathbb{A}))$

with $I_X^G(f) \neq 0$, the elements $X_v$ and $X'_v$ are conjugate under $G(\mathfrak{o}_v)$ for $v \notin S$; see [9, Lemma 7.2].[26] It follows that

$$\{X \in \Gamma_{X^*}(\mathfrak{g}(\mathbb{A})) \mid I_X^G(f) \neq 0\}$$

is finite, so the order of summation over $X$ and $\kappa$ may be interchanged. We find

$$J^G(f) = \tau(G^*) \sum_{X^* \in \Sigma_{\mathrm{ell}}(\mathfrak{g}^*)} \sum_{\kappa \in \mathcal{K}(T_{X^*}, G^*, F)} I_{X^*}^{G,\kappa}(f), \tag{10.3}$$

where

$$I_{X^*}^{G,\kappa}(f) = \sum_{X \in \Gamma_{X^*}(\mathfrak{g}(\mathbb{A}))} \langle \kappa, \mathrm{inv}(X^*, X)\rangle I_X^G(f).$$

**10.6.** In this paragraph, $F$ is local or global.

Start with an endoscopic triple $(H, \kappa, \xi)$ and an element $Y \in \mathfrak{h}_{G\text{-reg}}(F)$. There exists $X^* \in \Sigma(\mathfrak{g}^*)$ such that $Y \to X^*$. In §4.4, we explained how to define an element $\kappa_Y \in \mathcal{K}(T_{X^*}, G^*, F)$. We state the following lemma.

**Lemma 10.3.** *For every $X^* \in \Sigma_{\mathrm{ell}}(\mathfrak{g}^*)$ and $\kappa \in \mathcal{K}(T_{X^*}, G^*, F)$, there exist an elliptic endoscopic triple $(H, \kappa, \xi)$ and $Y \in \Sigma(\mathfrak{h})$ such that $Y \to X^*$ and $\kappa_Y = \kappa$.*

*Any two such data $(H, \kappa, \xi, Y)$ and $(H', \kappa', \xi', Y')$ are equivalent as endoscopic data for $G$ by an isomorphism carrying $Y$ to $Y'$. Such an isomorphism is unique up to an inner automorphism.*

*Proof.* We discuss only existence; a complete proof is given in [3, Lemma 9.7]. Let $X^* \in \mathfrak{g}^*_{\mathrm{ell}}(F)$, and set $T = T_{X^*}$. Let $\kappa \in \mathcal{K}(T_{X^*}, G^*, F)$. In particular, $\kappa$ defines a character of $H^1(C_F, T_{\mathrm{sc}})$. Fix a finite Galois extension $E$ of $F$ splitting $T$. Tate–Nakayama duality identifies $H^1(C_F, T_{\mathrm{sc}})$ with $\widehat{H}^{-1}(E/F, X_*(T_{\mathrm{sc}}))$. Since $X_*(T_{\mathrm{sc}})^\Gamma = 1$ by ellipticity, we also have

$$H^1(C_F, T_{\mathrm{sc}}) \simeq H_0(E/F, X_*(T_{\mathrm{sc}})).$$

Thus we may regard $\kappa$ as a character of $H_0(E/F, X_*(T_{\mathrm{sc}}))$. The root datum

$$\left(X^*(T_G), R_\kappa, X_*(T_G), R_\kappa^\vee\right)$$

is stable under $\Gamma$. Moreover, the action of $\Gamma$ carries a base to a base. This root datum is therefore the root datum of a group $H$. One readily completes these data to an endoscopic triple $(H, \kappa, \xi)$. The torus $T$ identifies with a subtorus of $H$, and this identification carries $X^*$ to a $G$-regular element $Y \in \mathfrak{h}(F)$. This constructs the quadruple $(H, \kappa, \xi, Y)$. □

**10.7. Normalization of transfer factors.** Let $F$ be a global field and $(H, \kappa, \xi)$ an endoscopic datum for $G$. Let $V$ be the set of places of $F$. For every $v \in V$, we obtain in the evident way a local endoscopic datum $(H_v, \kappa, \xi)$ for $G_v$, where $G_v$ denotes the group over the completion $F_v$ obtained by extending scalars.

Let $Y_i \in \mathfrak{h}_{G\text{-reg}}(F)$ and $X_i^* \in \mathfrak{g}^*_{\mathrm{reg}}(F)$ satisfy $Y_i \to X_i^*$ for $i = 1, 2$. Suppose that $\Gamma_{X_i^*}(\mathfrak{g}(\mathbb{A})) \neq \varnothing$, and choose $X_i \in \Gamma_{X_i^*}(\mathfrak{g}(\mathbb{A}))$.

[26] *Translator's note.* The source imposes nonvanishing only on $I_X^G(f)$ in this sentence; it does not state the corresponding condition on $X'$. The wording is retained.

At every place $v$, there is a relative transfer factor denoted by $\Delta_{G_v,H_v}$, or simply $\Delta_v$; see Section 7. One verifies the following formula; see [7, Section 6]:

$$\langle \kappa_{Y_1}, \mathrm{inv}(X_1^*, X_1)\rangle \langle \kappa_{Y_2}, \mathrm{inv}(X_2^*, X_2)\rangle^{-1} = \prod_{v\in V} \Delta_v(Y_1, X_1, Y_2, X_2),$$

where $\Delta_v(Y_1, X_1, Y_2, X_2) = 1$ for almost every $v$.

Normalize the transfer factor by taking $(Y_2, X_{2,v})$ as base pair.[27] We require

- $\Delta_v(Y_2, X_{2,v}) = 1$ for almost every $v$;
- $\prod_{v\in V} \Delta_v(Y_2, X_2) = \langle \kappa_{Y_2}, \mathrm{inv}(X_2^*, X_2)\rangle$.

Once the transfer factors are normalized in this way, these last two formulas hold for all $Y \in \mathfrak{h}_{G\text{-reg}}(F)$, $X^* \in \mathfrak{g}^*_{\mathrm{reg}}(F)$, and $X \in \Gamma_{X^*}(\mathfrak{g}(\mathbb{A}))$ such that $Y \to X^*$.

**10.8.** Return to the notation at the end of §10.5. Let $X^* \in \Sigma_{\mathrm{ell}}(\mathfrak{g}^*)$ and $\kappa \in \mathcal{K}(T_{X^*}, G^*, F)$. Let $(H, \kappa, \xi)$ be an endoscopic datum and $Y \in \mathfrak{h}_{G\text{-reg}}(F)$ such that $Y \to X^*$ and $\kappa_Y = \kappa$, as in Lemma 10.3. Suppose that $f = \bigotimes_{v\in V} f_v$. Using the preceding normalization of transfer factors, we find

$$I^{G,\kappa}_{X^*}(f) = \prod_{v\in V} I^{G_v,H_v}_Y(f_v),$$

where $I^{G_v,H_v}_Y(f_v) = 1$ for almost every $v \in V$; see [9, Section 7 and §10.2]. At this point one must establish compatibility relations between local and global Haar measures. We will not discuss them.

## 11. Waldspurger's result

**11.1.** Let $F$ be a global field and $u$ a finite place of $F$. Let $(H, \kappa, \xi)$ be an endoscopic datum. Assume the following conditions.

1. $F$ is totally imaginary.
2. For every $v \in V$, there exist $Y \in \mathfrak{h}_{G\text{-reg}}(F_v)$ and $X \in \mathfrak{g}_{\mathrm{reg}}(F_v)$ such that $Y \to X$.
3. The place $u$ is inert in $F[H]$, the smallest extension over which $H$ splits.
4. The datum $(H_u, \kappa, \xi)$ is elliptic.

Here is the theorem proved by Waldspurger in [9].

**Theorem 11.1.** *Let $w \in V$ be a finite place distinct from $u$. Then Theorem 8.4 holds for the pair $(G_w, H_w)$.*

From this, globalization or descent arguments show that Theorem 8.4 holds in full generality; see [9, Sections 4, 5, 6, and 11]. Note that proving Theorem 11.1 requires the fundamental lemma, Theorem 8.2, for almost every pair $(G_v, H_v)$.

[27] *Translator's note.* The source writes $(Y, X_{2,v})$ here and in the first normalization condition. We restore the subscript 2.

**11.2.** Fix a finite place $w \neq u$ of $F$. Let

$$\langle\ ,\ \rangle : \mathfrak{g}(F) \times \mathfrak{g}(F) \longrightarrow F$$

be a nondegenerate symmetric bilinear form invariant under $G(F)$, and let $\psi : \mathbb{A}_F/F \to \mathbb{C}^\times$ be a continuous nontrivial character. As in §8.8, this gives an analogous form on $\mathfrak{h}(F)$.

Fix two $\mathcal{O}_F$-lattices $\mathfrak{a} \subset \mathfrak{g}(F)$ and $\mathfrak{b} \subset \mathfrak{h}(F)$. They give $\mathcal{O}_{F_v}$-lattices $\mathfrak{a}_v \subset \mathfrak{g}(F_v)$ and $\mathfrak{b}_v \subset \mathfrak{h}(F_v)$.[28] There is a finite set $S_1$ of places containing the archimedean places and the places $u, w$ such that, for every $v \notin S_1$, the lattices $\mathfrak{a}_v$ and $\mathfrak{b}_v$ are hyperspecial and self-dual for the bicharacters $\psi(\langle\cdot,\cdot\rangle)$.

**11.3.** The set of equivalence classes of endoscopic data for the quasi-split inner form $G^*$ of $G$ that are unramified outside $S_1$ is finite; see [6, Lemma 8.12]. Fix a set $\mathcal{H}$ of representatives. Since $H$ is unramified outside $S_1$, we may, and do, assume that $(H, \kappa, \xi) \in \mathcal{H}$. Let $\mathcal{H}' \subset \mathcal{H}$ be the subset consisting of data $(H', \kappa', \xi')$ such that $F[H'] \not\subset F[H]$, that is, such that $H'$ is not split over $F[H]$. Choose

$$v = v_{(H',\kappa',\xi')} \in V \setminus S_1$$

such that its Frobenius in $\mathrm{Gal}(F[H]F[H']/F)$ fixes $F[H]$ but not $F[H']$. Then $H_v$ is split, whereas $H'_v$ is not. Set

$$S_2 = \{v_{(H',\kappa',\xi')} \mid (H', \kappa', \xi') \in \mathcal{H}'\}, \qquad S = S_1 \cup S_2.$$

**11.4.** We wish to prove that there is a constant $c \neq 0$ with the following property. For every $f_w \in C_c^\infty(\mathfrak{g}(F_w))$, every transfer $f_w^H \in C_c^\infty(\mathfrak{h}(F_w))$ of $f_w$, in the sense of Theorem 8.1, and every $Y_w \in \Sigma_G(\mathfrak{h})$, one has[29]

$$c\, \widehat{I}_{Y_w}^{G_w,H_w}(f_w) = \widehat{S}_{Y_w}^{H_w}(f_w^H). \tag{11.1}$$

In fact, we shall see that

$$c = \prod_{\substack{v \in S \setminus V_\infty \\ v \neq w}} c_v^{-1},$$

where $c_v$ is the constant introduced in condition 7 of Proposition 11.2.

Let $Y_w \in \mathfrak{h}_{G\text{-reg}}(F_w)$. Fix a compact open subset $\Omega_w \subset \mathfrak{h}_{G\text{-reg}}(F_w)$ containing $Y_w$ on which both sides of the desired identity are constant.

**11.5.** Let $V_\infty$ be the set of archimedean places. For every $v \in S \setminus V_\infty$ with $v \neq w$, fix a compact open subset $\Omega_v \subset \mathfrak{h}_{G\text{-reg}}(F_v)$ such that

1. every point of $\Omega_v$ is an image of an element of $\mathfrak{g}_{\mathrm{reg}}(F_v)$; this is possible by condition 2 of §11.1;
2. if $v \in S_2$, every point of $\Omega_v$ has a split centralizer torus; this is possible because $H_v$ is split at such a place;
3. every point of $\Omega_u$ is elliptic; this is compatible with condition 1 by condition 4 of §11.1 and Lemma 4.6.

[28] *Translator's note.* The source writes $\mathfrak{g}(F)$ and $\mathfrak{h}(F)$ for the ambient spaces of the localized lattices. The completions $F_v$ are made explicit here; the correction to the target of the bilinear form was noted in §5.4.

[29] *Translator's note.* The source writes $f$, rather than $f_w$, on the left of this local identity.

**11.6.** For every maximal subtorus $T$ of $G^*$ defined over $F$, there is a natural map

$$\mathcal{E}(T, G^*, F_v) \longrightarrow \mathcal{E}(T, G^*, \mathbb{A}_F/F),$$

and hence a dual map

$$\mathcal{K}(T, G^*, \mathbb{A}_F/F) \longrightarrow \mathcal{K}(T, G^*, F_v),$$

denoted by $\kappa \mapsto \kappa_v$.[30]

**11.7.** Choose $Y_0 \in \mathfrak{h}(F)$ such that

1. $Y_0 \in \Omega_v$ for every $v \in S \setminus V_\infty$;
2. $Y_0 \in \mathfrak{b}_v$ for every $v \in V \setminus S$.

Strong approximation guarantees the existence of $Y_0$. Let $X_0^* \in \mathfrak{g}^*_{\mathrm{reg}}(F)$ satisfy $Y_0 \to X_0^*$. One checks that $X_0^*$ is regular semisimple and elliptic. Let $\mathcal{Y} \subset \Sigma_G(\mathfrak{h})$ be the finite subset of classes $Y$ such that $Y \to X_0^*$, identified with a set of representatives. Of course, we assume that $Y_0 \in \mathcal{Y}$. At the end of §4.4, we explained how every element of $\mathcal{Y}$ defines an element $\kappa_Y \in \mathcal{K}(T_{X_0^*}, G^*, F)$. Set

$$\kappa_0 = \kappa_{Y_0}, \qquad \mathcal{Y}_0 = \{Y \in \mathcal{Y} \mid \kappa_Y = \kappa_0\}.$$

**11.8. Choice of functions outside $S$.** For every $v \notin S$, let $f_v$, respectively $f_v^H$, be the characteristic function of $\mathfrak{a}_v$, respectively $\mathfrak{b}_v$. Let $c_v \neq 0$ be the constant supplied by Theorem 8.2.

**11.9. Construction of functions at finite places.** Let $v \in S$ be a finite place distinct from $w$. Let $X^* \in \mathfrak{g}^*_{\mathrm{reg}}(F_v)$. Set[31]

$$\mathfrak{g}(X^*, F_v) = \chi_G^{-1}(\chi_{G^*}(X^*)) \cap \mathfrak{g}(F_v).$$

Let $\kappa \in \mathcal{K}(T_{X^*}, G^*, F_v)$. A *$\kappa$-function above $X^*$* is a function $\theta : \mathfrak{g}(X^*, F_v) \to \mathbb{C}$ such that[32]

– $\theta$ is $G(F_v)$-invariant;
– for $X_1, X_2 \in \mathfrak{g}(X^*, F_v)$,

$$\theta(X_1) = \theta(X_2)\langle \kappa, \mathrm{inv}(X^*; X_1, X_2)\rangle.$$

For such a function $\theta$ and $f \in C_c^\infty(\mathfrak{g}(F_v))$, set

$$I^\theta_{X^*}(f) = \sum_X \theta(X) I_X(f),$$

where the sum is over $X \in \mathfrak{g}(X^*, F_v)$ modulo conjugacy. When $\theta = \Delta(Y, \cdot)$, as in (7.2) of §7.4, for $Y \in \mathfrak{h}(F_v)$ such that $Y \to X^*$, we have

$$I^\theta_{X^*}(f) = I_Y^{G,H}(f).$$

[30] *Translator's note.* Here and later the source writes $\mathcal{K}(T, G^*, \mathbb{A}_F/F)$ for the global character group denoted by $\mathcal{K}(T, G^*, F)$ in §2.6. This variation in notation is retained.

[31] *Translator's note.* The source writes $\mathfrak{g}_{\mathrm{reg}}(F_v)$ rather than $\mathfrak{g}^*_{\mathrm{reg}}(F_v)$ for $X^*$, and suppresses the subscript $v$ in several occurrences of $F$ in this local construction. We use $F_v$ consistently in §11.9.

[32] *Translator's note.* The source writes $\kappa_Y$ in this definition, although no $Y$ has been chosen. We write the character $\kappa$ being defined.

When $G = G^*$ and $\varphi$ is the identity, we can take

$$\theta = \langle \kappa, \mathrm{inv}(X^*; \cdot, X^*) \rangle,$$

and then write $I^{\kappa}_{X^*}(f) = I^{\theta}_{X^*}(f)$.[33]

We now come to the following construction of Waldspurger; see [9, Proposition 8.2]. Notation not recalled here is as in the preceding paragraphs.

**Proposition 11.2.** *There exist functions $f \in C_c^\infty(\mathfrak{g}_{\mathrm{reg}}(F_v))$ and $f^H \in C_c^\infty(\mathfrak{h}_{G\text{-reg}}(F_v))$ with the following properties.*[34]

1. *If $X \in \mathrm{supp}(f)$, respectively $Y \in \mathrm{supp}(f^H)$, there exists $X' \in \mathfrak{t}_{X_0^*}(F_v)$ such that $\chi_{G^*}(X') = \chi_G(X)$, respectively such that $Y \to X'$.*
2. *$f^H$ is a transfer of $f$.*
3. *Let $Z \in \mathfrak{g}^*_{\mathrm{reg}}(F_v)$ and $\tau \in \mathcal{K}(T_Z, G^*, F_v)$. Suppose that there is no $g \in G^*$ such that conjugation by $g$ induces an $F_v$-isomorphism from $(T_{X_0^*}, \kappa_0)$ to $(T_Z, \tau)$. Then, for every $\tau$-function $\theta$ above $Z$,*
$$I^{\theta}_Z(f) = 0.$$
4. *Let $Y \in \mathfrak{h}_{G^*\text{-reg}}(F_v)$ and $\tau \in \mathcal{K}(T_Y, H, F_v)$. If $\tau \neq 1$, then*
$$I^{\tau}_Y(f^H) = 0.$$
5. *Let $\tau \in \mathcal{K}(T_{X_0^*}, G^*, F_v)$ and let $\theta$ be a nonzero $\tau$-function above $X_0^*$. For $\tau \neq \kappa_0$, one has*
$$I^{\tau}_{X_0^*}(\widehat{f}) = 0.$$
*Moreover,*
$$I^{\kappa_0}_{X_0^*}(\widehat{f}) \neq 0.$$
6. *Let $Y \in \mathcal{Y}$ and $\tau \in \mathcal{K}(T_Y, H, F_v)$. Then*
$$I^{\tau}_Y(\widehat{f^H}) = 0$$
*whenever $Y \notin \mathcal{Y}_0$ or $\tau \neq 1$. Moreover,*
$$I^{G_v, H_v}_Y(\widehat{f}) = 0$$
*if $Y \notin \mathcal{Y}_0$.*
7. *There is a constant $c_v \neq 0$, depending only on $(G_v, H_v, \psi_v, \langle\ ,\ \rangle)$, such that, for every $Y \in \mathcal{Y}_0$,*
$$S^{H_v}_Y(\widehat{f^H}) = c_v I^{G_v, H_v}_Y(\widehat{f}) \neq 0.$$

We will not give the elaborate construction of these functions here. The reader can check that it is easy to construct pairs satisfying conditions 1–4. Conditions 5–7 are much harder to arrange. Their construction rests on an explicit expression "at infinity" for the functions $\widehat{i}^G$ of the theorem. More precisely, let $X, Z \in \mathfrak{g}_{\mathrm{reg}}(F_v)$. Waldspurger shows

[33] *Translator's note.* The source writes $\theta = \mathrm{inv}(X^*; \cdot, X^*)$ without evaluating the cohomology class against $\kappa$. The pairing is supplied to obtain the required complex-valued function.

[34] *Translator's note.* In condition 5, the source introduces a nonzero $\theta$ but uses the symbols $I^{\tau}$ and $I^{\kappa_0}$, whose earlier explicit definition assumed $G = G^*$. This notational ambiguity is retained rather than replaced by a new normalization. In condition 6, the source's otherwise undefined $J^{\tau}_Y$ is written as $I^{\tau}_Y$, consistently with the surrounding formulas. The localized character $\kappa_{0,v}$ is abbreviated to $\kappa_0$ in the proposition, as in the source.

that there is a nonnegative integer $N$ such that, for every $\lambda \in F_v^\times$ with $v(\lambda) < -N$, the quantity $\widehat{i}^G(\lambda X, Z)$ has an explicit expression; see [8, Proposition VIII.1].

**11.10.** For every finite place $v \in S$ distinct from $w$, choose $f_v \in C_c^\infty(\mathfrak{g}_{\mathrm{reg}}(F_v))$ and $f_v^H \in C_c^\infty(\mathfrak{h}_{G\text{-reg}}(F_v))$ satisfying the conclusions of Proposition 11.2.

**11.11.** Let $v \in V_\infty$ be an archimedean place. Recall that we assume $F_v \simeq \mathbb{C}$. Let $f_v \in \mathcal{S}(\mathfrak{g}(F_v))$ be arbitrary, and let $f_v^H \in \mathcal{S}(\mathfrak{h}(F_v))$ be a transfer of $f_v$, as in Theorem 9.2.

**11.12.** Set $f = \bigotimes_{v\in V} f_v$ and $f^H = \bigotimes_{v\in V} f_v^H$. Let $\mathrm{Out}_G(H)$ be the group of automorphisms of the datum $(H, \kappa, \xi)$ modulo inner automorphisms. It is a finite group. The following proposition is Proposition 10.9 of [9]. Set

$$c^S = \prod_{v\notin S} c_v,$$

where $c_v$ is defined in §11.8.

**Proposition 11.3.** *One has*

$$J^H(f^H) = |\,\mathrm{Out}_G(H)| c^S \tau(H)\tau(G^*)^{-1} J^G(f).$$

*Proof.* Start with (10.3) in §10.5:

$$J^G(f) = \tau(G^*) \sum_{X^*\in\Sigma_{\mathrm{ell}}(\mathfrak{g}^*)} \sum_{\kappa\in\mathcal{K}(T_{X^*},G^*,F)} I_{X^*}^{G,\kappa}(f).$$

Let $X^* \in \Sigma_{\mathrm{ell}}(\mathfrak{g}^*)$ and $\kappa \in \mathcal{K}(T_{X^*}, G^*, F)$. Let $(H', \kappa', \xi')$ be an elliptic endoscopic datum for $G^*$ and $Y \in \Sigma(\mathfrak{h}')$ such that $Y \to X^*$ and $\kappa_Y = \kappa$, as in Lemma 10.3. By §10.8,

$$I_{X^*}^{G,\kappa}(f) = \prod_{v\in V} I_Y^{G_v,H'_v}(f_v).$$

As usual, almost all factors in the infinite products under consideration are equal to 1. In what follows, suppose that $I_{X^*}^{G,\kappa}(f) \neq 0$.

This implies that $I_Y^{G_v,H'_v}(f_v) \neq 0$ for every $v \notin S$. But $f_v$ is the characteristic function of a hyperspecial lattice. It follows that $H'_v$ is unramified; this is a version of a lemma of Kottwitz, see [9, Lemma 7.4]. Furthermore, for $v \in S_2$, the nonvanishing of $I_Y^{G_v,H'_v}(f_v)$ implies that the centralizer of $Y$ is split, by condition 1 of Proposition 11.2 and condition 2 of §11.5. Thus $H'$ is unramified outside $S_1$.[35]

We may therefore assume that $(H', \kappa', \xi') \in \mathcal{H}$. If it lies in $\mathcal{H}'$, then, at $v = v_{(H',\kappa',\xi')}$, the nonvanishing of $I_Y^{G_v,H'_v}(f_v)$ implies as above that $H'_v$ is split, a contradiction. Hence $F[H'] \subset F[H]$.

We now use the place $u$. By condition 3 of Proposition 11.2 and Lemma 10.3, the data $(H_u, \kappa, \xi)$ and $(H'_u, \kappa', \xi')$ are equivalent over $F_u$. Condition 3 of §11.1 shows that this isomorphism extends to an $F$-isomorphism.

[35] *Translator's note.* The source writes $H$ in this last sentence. The argument concerns $H'$.

Let $\Sigma_G(\mathfrak{h})(X^*,\kappa)$ be the subset of $\Sigma_G(\mathfrak{h})$ consisting of $Y$ such that $Y \to X^*$ and $\kappa_Y = \kappa$. If nonempty, this subset has cardinality $|\operatorname{Out}_G(H)|$, by Lemma 10.3. It follows that[36]

$$J^G(f) = \frac{\tau(G^*)}{|\operatorname{Out}_G(H)|} \sum_{X^* \in \Sigma_{\mathrm{ell}}(\mathfrak{g}^*)} \sum_{\kappa \in \mathcal{K}(T_{X^*}, G^*, F)} \cdot \sum_{Y \in \Sigma_G(\mathfrak{h})(X^*,\kappa)} \prod_{v \in V} I_Y^{G_v, H_v}(f_v).$$

We make the analogous calculation for $H$. Since the support of $f_u^H$ consists of $G$-regular elements, we may write[37]

$$J^H(f^H) = \tau(H) \sum_{Y \in \Sigma_{G,\mathrm{ell}}(\mathfrak{h})} \sum_{\kappa \in \mathcal{K}(T_Y, H, F)} I_Y^{H,\kappa}(f^H).$$

Let $Y \in \Sigma_{G,\mathrm{ell}}(\mathfrak{h})$ and $\kappa \in \mathcal{K}(T_Y, H, F)$. In §11.6, we defined $\kappa_v \in \mathcal{K}(T_Y, H, F_v)$ for every place $v$. One checks that

$$I_Y^{H,\kappa}(f^H) = \prod_{v \in V} I_Y^{H,\kappa_v}(f_v^H).$$

If this product is nonzero, then $I_Y^{H,\kappa_u}(f_u^H) \neq 0$, which forces $\kappa_u = 1$ by condition 4 of Proposition 11.2. Since $T_Y$ is $F$-elliptic and even $F_u$-elliptic, by the construction of $f_u$ and condition 3 of §11.5, one readily sees, for example using Tate–Nakayama duality, that

$$\mathcal{K}(T_Y, H, \mathbb{A}_F/F) \longrightarrow \mathcal{K}(T_Y, H, F_u)$$

is injective. Consequently $\kappa = 1$. We obtain

$$J^H(f^H) = \tau(H) \sum_{Y \in \Sigma_{G,\mathrm{ell}}(\mathfrak{h})} \prod_{v \in V} S_Y^{H_v}(f_v^H).$$

Using the transfer properties of $f_v$ and $f_v^H$, and in particular the fundamental lemma, we write

$$J^H(f^H) = \tau(H) c^S \sum_{Y \in \Sigma_{G,\mathrm{ell}}(\mathfrak{h})} \prod_{v \in V} I_Y^{G_v, H_v}(f_v).$$

To conclude, it is enough to use the fact that $\Sigma_{G,\mathrm{ell}}(\mathfrak{h})$ is the disjoint union of the sets $\Sigma_{G,\mathrm{ell}}(\mathfrak{h})(X^*,\kappa)$ as $X^* \in \Sigma_{\mathrm{ell}}(\mathfrak{g}^*)$ and $\kappa \in \mathcal{K}(T_{X^*}, G^*, F)$ vary. □

**11.13. Refining the conditions at the archimedean places.** We may further assume, by [9, Lemma 10.7], that the archimedean functions are chosen so that

1. for every $v \in V_\infty$,
$$I_{\varphi_v^{-1}(X_0^*)}(\widehat{f_v}) \neq 0;$$
2. if $X \in \mathfrak{g}^*(F)$ and, at every place $v$, $\varphi_v^{-1}(X_v)$ is conjugate under $G(\overline{F_v})$ to an element of $\operatorname{supp}(\widehat{f_v})$, then $X$ is stably conjugate to $X_0^*$;
3. if $Y \in \mathfrak{h}(F)$ and, at every place $v$, $Y$ is conjugate under $H(\overline{F_v})$ to an element of $\operatorname{supp}(\widehat{f_v^H})$, then $Y \to X_0^*$.

[36] *Translator's note.* The factor $\tau(G^*)$ is absent from this display in the source, although it appears in the starting formula and is needed for the stated proposition. It is restored here.

[37] *Translator's note.* The source writes $J^G(f)$ on the left of this formula for the group $H$.

Of course, at the archimedean places we continue to require $f_v^H$ to be a transfer of $f_v$, so that $\widehat{f_v^H}$ is a transfer of $\widehat{f}_v$, by Theorem 9.2.[38]

**11.14. Using the Poisson summation formula.** The formula gives the following identity for every $x \in G(\mathbb{A})$:

$$\sum_{X \in \mathfrak{g}(F)} f(\mathrm{Ad}(x^{-1})X) = \sum_{X \in \mathfrak{g}(F)} \widehat{f}(\mathrm{Ad}(x^{-1})X).$$

The shape of the support of $f_u$ shows that the left-hand sum is effectively over $X \in \mathfrak{g}_{\mathrm{ell}}(F)$. By the preceding paragraph, the same is true of the right-hand sum. Integrating over $G(F)\backslash G(\mathbb{A})^1$, using Theorem 10.1, gives

$$J^G(f) = J^G(\widehat{f}). \tag{11.2}$$

The same reasoning gives

$$J^H(f^H) = J^H(\widehat{f^H}). \tag{11.3}$$

**11.15. Calculation of $J^G(\widehat{f})$.** We have the following lemma.

**Lemma 11.4.** *One has*

$$J^G(\widehat{f}) = \tau(G^*) I^{\kappa_0}_{X_0^*}(\widehat{f}).$$

*Proof.* Apply pre-stabilization to $J^G(\widehat{f})$. Since we also have condition 2 of §11.13, we obtain

$$J^G(\widehat{f}) = \tau(G^*) \sum_{\kappa \in \mathcal{K}(T_{X_0^*}, G^*, F)} I^{\kappa}_{X_0^*}(\widehat{f}).$$

Let $\kappa \in \mathcal{K}(T_{X_0^*}, G^*, F)$ satisfy $I^{\kappa}_{X_0^*}(\widehat{f}) \neq 0$. Choose $(H', \kappa', \xi')$ and $Y \in \mathfrak{h}'_{G\text{-reg}}(F)$ such that $Y \to X_0^*$ and $\kappa'_Y = \kappa$, as in Lemma 10.3. Then

$$I^{\kappa}_{X_0^*}(\widehat{f}) = \prod_{v \in V} I_Y^{G_v, H'_v}(\widehat{f}_v).$$

Since $I_Y^{G_u, H'_u}(\widehat{f}_u) \neq 0$, condition 5 of Proposition 11.2 gives $\kappa_{0,u} = \kappa_u$. As $T_{X_0^*}$ is both $F$-elliptic and $F_u$-elliptic, the localization map for endoscopic characters is injective, as explained above.[39] Hence $\kappa_0 = \kappa$, as required. □

**11.16. Calculation of $J^H(\widehat{f^H})$.** We have the following lemma.

**Lemma 11.5.** *One has*

$$J^H(\widehat{f^H}) = \tau(H) \sum_{Y \in \mathcal{Y}_0} \prod_{v \in V} S_Y^{H_v}(\widehat{f_v^H}).$$

[38] *Translator's note.* The source suppresses the subscripts $v$ in this last sentence. They are restored to make clear that the statement concerns the archimedean choices of this paragraph, not the Fourier compatibility at $w$ that is being proved.

[39] *Translator's note.* The source says $F_v$-elliptic here. The place used in this argument is $u$.

*Proof.* Apply pre-stabilization to $J^H(\widehat{f^H})$. This gives[40]

$$J^H(\widehat{f^H}) = \tau(H) \sum_{Y \in \mathcal{Y}} \sum_{\kappa \in \mathcal{K}(T_Y, H, F)} I_Y^\kappa(\widehat{f^H}).$$

Let $Y \in \mathcal{Y}$ and $\kappa \in \mathcal{K}(T_Y, H, F)$ satisfy $I_Y^\kappa(\widehat{f^H}) \neq 0$. We have[41]

$$I_Y^\kappa(\widehat{f^H}) = \prod_{v \in V} I_Y^{\kappa_v}(\widehat{f_v^H}).$$

By condition 6 of Proposition 11.2, $Y \in \mathcal{Y}_0$ and $\kappa_u = 1$. The same injectivity argument gives $\kappa = 1$. The factors in the product are therefore stable orbital integrals. □

**11.17. Conclusion.** Lemmas 11.4 and 11.5, Proposition 11.3, and identities (11.2) and (11.3) give

$$\sum_{Y \in \mathcal{Y}_0} \prod_{v \in V} S_Y^{H_v}(\widehat{f_v^H}) = c^S |\operatorname{Out}_G(H)| I_{X_0^*}^{\kappa_0}(\widehat{f}). \tag{11.4}$$

Set[42]

$$c^w = \prod_{\substack{v \in V \setminus V_\infty \\ v \neq w}} c_v.$$

Using the fundamental lemma, the properties of $f_v$ at finite places $v \in S$ distinct from $w$, especially condition 7 of Proposition 11.2, and the properties at archimedean places, we obtain, for every $Y \in \mathcal{Y}_0$,

$$\prod_{v \in V} S_Y^{H_v}(\widehat{f_v^H}) = c^w S_Y^{H_w}(\widehat{f_w^H}) \prod_{\substack{v \in V \\ v \neq w}} I_Y^{G_v, H_v}(\widehat{f}_v).$$

Let $v \in V$ with $v \neq w$. Choose $X_0 \in \mathfrak{g}(F_v)$ such that $Y \to X_0$. Equivalently, $X_0$ and $X_0^*$ have the same image in $\mathfrak{c}$. In particular, this condition does not depend on the choice of $Y \in \mathcal{Y}_0$.[43] For every $X \in \mathfrak{g}(F_v)$ such that $Y \to X$, formula (7.2) in §7.4 gives[44]

$$\begin{aligned} \Delta_v(Y, X) &= \Delta_v(Y, X_0) \langle \kappa_{Y,v}, \operatorname{inv}(X_0^*; X, X_0) \rangle \\ &= \Delta_v(Y, X_0) \langle \kappa_{0,v}, \operatorname{inv}(X_0^*; X, X_0) \rangle \\ &= \frac{\Delta_v(Y, X_0)}{\Delta_v(Y_0, X_0)} \Delta_v(Y_0, X). \end{aligned}$$

Consequently,

$$I_Y^{G_v, H_v}(\widehat{f}_v) = \frac{\Delta_v(Y, X_0)}{\Delta_v(Y_0, X_0)} I_{Y_0}^{G_v, H_v}(\widehat{f}_v).$$

[40] *Translator's note.* The source omits $\tau(H)$ from this display, although it is present in pre-stabilization and in the statement of the lemma. It is restored here.

[41] *Translator's note.* In this product the source writes $\kappa_{Y,v}$ instead of the localization $\kappa_v$ of the character $\kappa$ just chosen.

[42] *Translator's note.* The source has $c_s$ rather than $c_v$ as the factor in this product.

[43] *Translator's note.* The source introduces $X \in \mathfrak{g}(F_v)$ but immediately imposes a condition on $X_0$ and then uses $X_0$ as base point. We explicitly name the base point $X_0$ here. As in the source, local subscripts on this base point are suppressed; its occurrence in a factor at $w$ means the corresponding local choice at $w$.

[44] *Translator's note.* The source writes $X^*$ inside these invariants. We use the fixed element $X_0^*$ of this paragraph.

Using the product formula for transfer factors in §10.7, we obtain

$$\prod_{\substack{v\in V\\ v\neq w}} I_Y^{G_v,H_v}(\widehat{f}_v) = \frac{\Delta_w(Y_0,X_0)}{\Delta_w(Y,X_0)} \prod_{\substack{v\in V\\ v\neq w}} I_{Y_0}^{G_v,H_v}(\widehat{f}_v).$$

**Lemma 11.6.** *For every $Y \in \mathcal{Y}_0$, one has*

$$S_{Y_0}^{H_w}(\widehat{f_w^H}) = \frac{\Delta_w(Y_0,X_0)}{\Delta_w(Y,X_0)} S_Y^{H_w}(\widehat{f_w^H}).$$

*Proof.* We sketch the proof; for further details, see [9, Section 3]. Since $\kappa_Y = \kappa_{Y_0}$, there is an automorphism $i$ of $(H,\kappa,\xi)$ carrying $Y_0$ to $Y$, by Lemma 10.3. One shows, as in [9, Lemma 2.4], that there is a constant $\delta \neq 0$ such that, for every $Z \in \mathfrak{h}_{G\text{-reg}}(F_w)$ and every $X \in \mathfrak{g}(F_w)$ with $Z \to X$,

$$\Delta_w(i(Z),X) = \delta \Delta_w(Z,X). \tag{11.5}$$

In particular, since $f_w^H$ is a transfer of $f_w$, for every $Z \in \mathfrak{h}_{G\text{-reg}}(F_w)$ we obtain

$$S_{i(Z)}^{H_w}(f_w^H) = \delta S_Z^{H_w}(f_w^H).$$

Next observe that $i$ preserves the form $\langle\cdot,\cdot\rangle$ on $\mathfrak{h}(F_w)$. Thus it also preserves the functions $\widehat{i}^{H_w}$. Using Theorem 6.3, we conclude that

$$S_{i(Z)}^{H_w}(\widehat{f_w^H}) = \delta S_Z^{H_w}(\widehat{f_w^H})$$

for every $Z \in \mathfrak{h}_{G\text{-reg}}(F_w)$. To finish, evaluate $\delta$ by taking $(Z,X) = (Y_0,X_0)$ in (11.5). □

By the preceding lemma,

$$\prod_{v\in V} S_Y^{H_v}(\widehat{f_v^H}) = c^w S_{Y_0}^{H_w}(\widehat{f_w^H}) \prod_{\substack{v\in V\\ v\neq w}} I_{Y_0}^{G_v,H_v}(\widehat{f}_v).$$

In particular, this product does not depend on $Y$.[45] Identity (11.4) becomes

$$c^w S_{Y_0}^{H_w}(\widehat{f_w^H}) \prod_{\substack{v\in V\\ v\neq w}} I_{Y_0}^{G_v,H_v}(\widehat{f}_v) = c^S I_{X_0^*}^{\kappa_0}(\widehat{f}).$$

Since we also have

$$I_{X_0^*}^{\kappa_0}(\widehat{f}) = \prod_{v\in V} I_{Y_0}^{G_v,H_v}(\widehat{f}_v),$$

it follows that

$$\Big(c^w S_{Y_0}^{H_w}(\widehat{f_w^H}) - c^S I_{Y_0}^{G_w,H_w}(\widehat{f}_w)\Big) \cdot \prod_{\substack{v\in V\\ v\neq w}} I_{Y_0}^{G_v,H_v}(\widehat{f}_v) = 0.$$

The choices were made precisely to ensure that the second factor is nonzero. Hence

$$c^w S_{Y_0}^{H_w}(\widehat{f_w^H}) = c^S I_{Y_0}^{G_w,H_w}(\widehat{f}_w).$$

This equality also holds at $Y_w$, by the choice of $\Omega_w$ in §11.4, proving (11.1).

[45] *Translator's note.* The source says that the product does not depend on $Y_0$; the variable being summed over in (11.4) is $Y \in \mathcal{Y}_0$.

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